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Artificial Intelligence - A.A. 2026/2027
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1 – Introduction to Artificial Intelligence

Prof. Fabio Persia

University of L'Aquila, DISIM Department

fabio.persia@univaq.it https://www.disim.univaq.it/FabioPersia.html

UNIVERSITÀ DEGLI STUDI DELL'AQUILA

DISIM Dipartimento di Ingegneria e Scienze dell'Informazione e Matematica

1- Introduction to Artificial Intelligence - Outline

- What AI is and why AI
- Today's pervasiveness of AI: examples, applications, adoption in industry, job market
- History of AI
- Artificial general intelligence (strong AI) and narrow AI (weak AI)
- Symbolic AI, knowledge representation, automated reasoning
- Intelligent agents and agent-centric AI
- Machine learning, deep learning, connectionist AI
- Neuro-symbolic AI
- AI goals and subfields
- Ethics of AI

1- Introduction to Artificial Intelligence
Materials for this lesson

[BOOK] Russell, Norvig, 2021 - Artificial Intelligence: A Modern Approach, 4th ed. (https://aima.cs.berkeley.edu/global-index.html), Chapter 1

[BOOK] Aggarwal, 2021 - Artificial Intelligence: A Textbook (http://www.charuaggarwal.net), Chapter 1

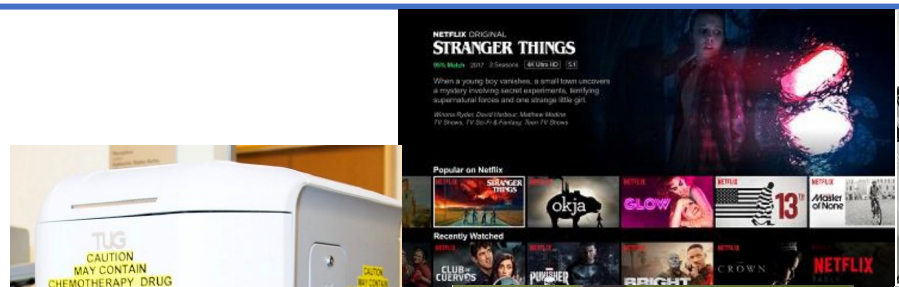
[COURSE] Introduction to Artificial Intelligence @UCBerkeley (https://inst.eecs.berkeley.edu/~cs188/su24/), Lecture 1

Artificial Intelligence (AI)

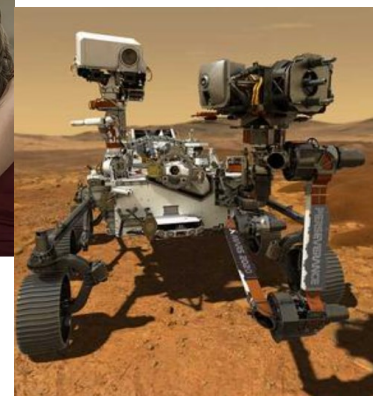
Artificial Intelligence (AI) is the field of study which focuses on computing systems able to imitate intelligent human behavior.

In simple terms, a computing system is said to exhibit AI when it can perform tasks that usually require human thinking and reasoning.

- It's like giving computing systems the ability to be smart like humans.

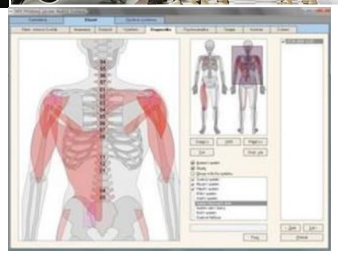


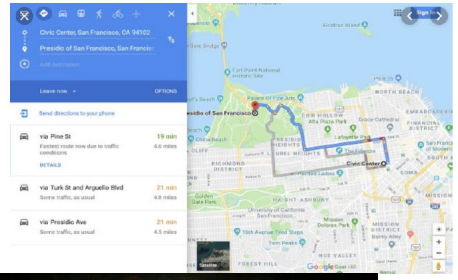


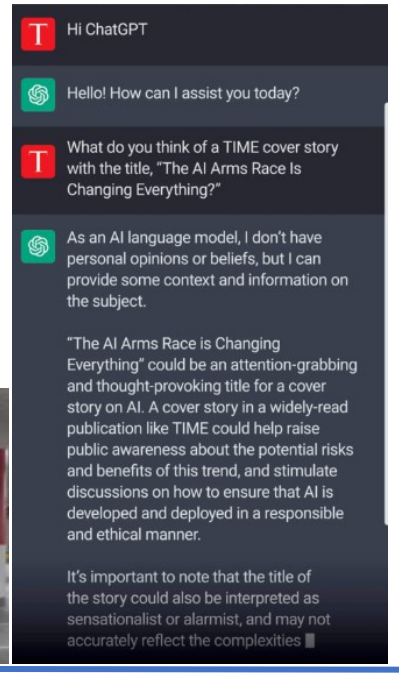




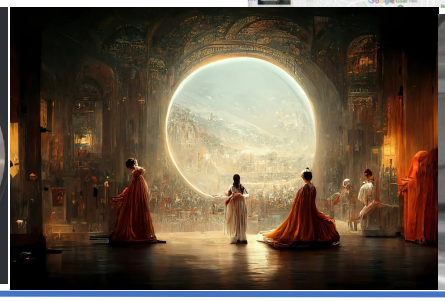














Intelligence

"Intelligence has been defined in many ways: the capacity for abstraction, logic, understanding, self-awareness, learning, emotional knowledge, reasoning, planning, creativity, critical thinking, and problem-solving. It can be described as the ability to perceive or infer information; and to retain it as knowledge to be applied to adaptive behaviors within an environment or context. (…)"

source: Wikipedia

Artificial Intelligence (AI)

AI is concerned with not just understanding but also building intelligent systems.

AI is a broad and multidisciplinary field that integrates concepts and techniques from various scientific and engineering domains, including

- philosophy, mathematics, economics, neuroscience, psychology, computer science, computer engineering, control theory, cybernetics, logistics

AI nowadays

AI boom

We live in the era of the so-called AI boom, or AI spring, an ongoing period of rapid progress in the field of artificial intelligence (AI) that started roughly in the late 2010s before gaining international prominence in the early 2020s.



image source

Among the key enablers of AI boom are:

- the advent of deep learning (starting form the late 2010s)
- the progresses in generative AI and large language models, fueled by the diffusion of tools like ChatGPT (starting form the early 2020s)

AI boom

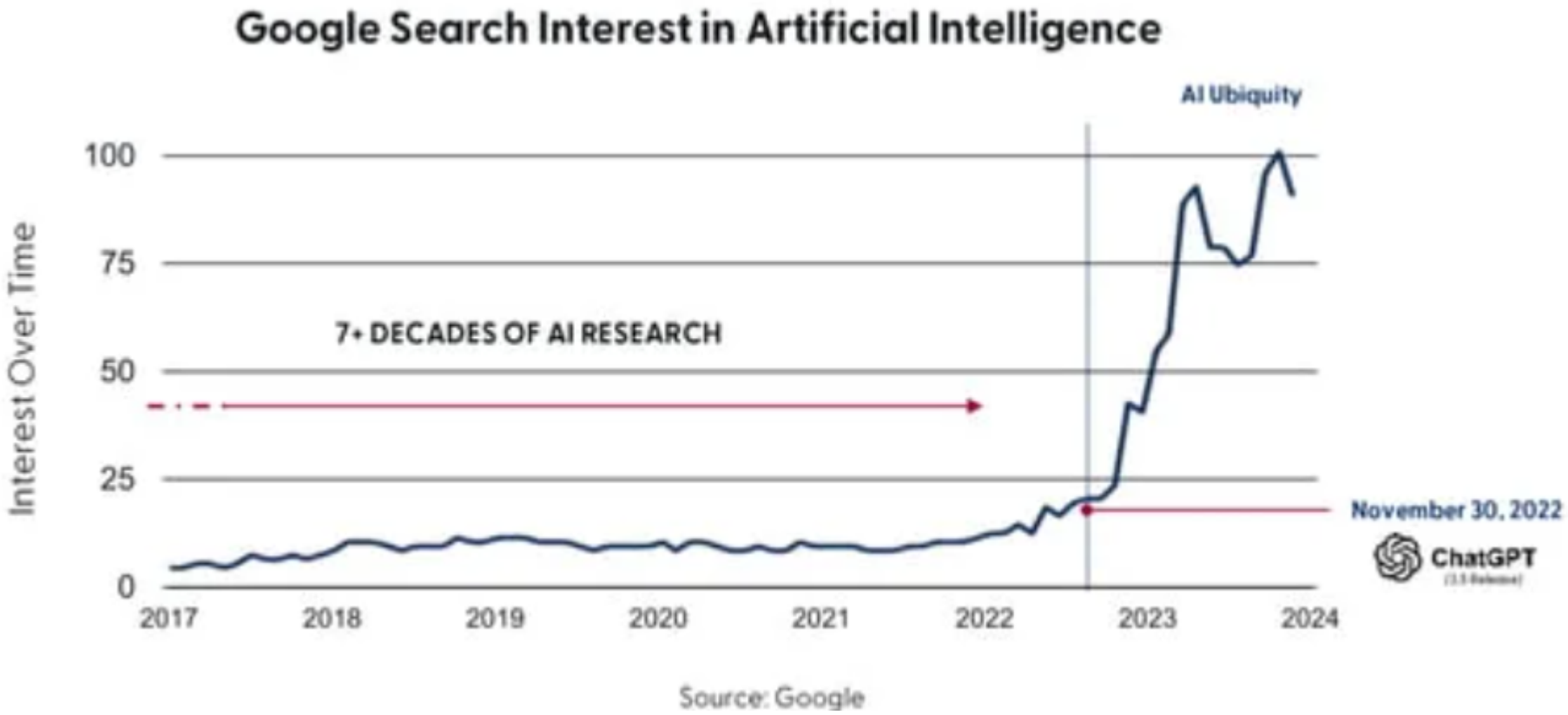


image source

Are we already here?

“Star Wars”, 1977



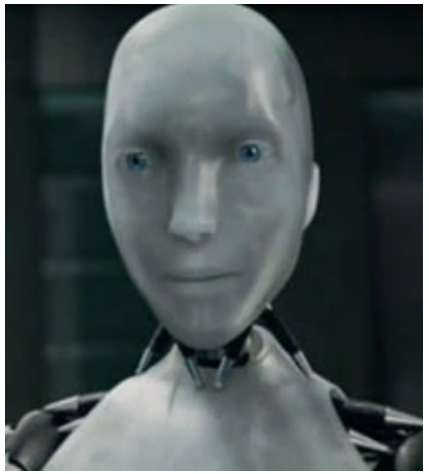
“Blade Runner”, 1982



“Matrix”, 1999



“I, Robot”, 2004



“Terminator”, 1984



“A.I. – Intelligenza Artificiale”, 2001



Robotic vehicles

In 2018, Waymo test vehicles passed the landmark of 10 million miles driven on public roads without a serious accident, with the human driver stepping in to take over control only once every 6,000 miles.



image source

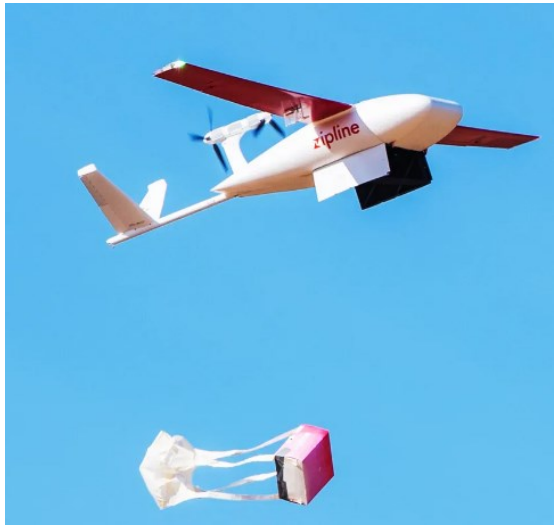


image source

In the air, autonomous fixed-wing drones have been providing cross-country blood deliveries in Rwanda since 2016. Quadcopters perform remarkable aerobatic maneuvers, explore buildings while constructing 3-D maps, and self-assemble into autonomous formations.

Robot vehicles



image source

Autonomous vacuum cleaners



source

Autonomous planning and scheduling

A hundred million miles from Earth, NASA's Remote Agent program became the first on-board autonomous planning program to control the scheduling of operations for a spacecraft (Jonsson et al., 2000). Remote Agent generated plans from high-level goals specified from the ground and monitored the execution of those plans—detecting, diagnosing, and recovering from problems as they occurred.

Today, the EUROPA planning toolkit (Barreiro et al., 2012) is used for daily operations of NASA's Mars rovers and the SEXTANT system (Winternitz, 2017) allows autonomous navigation in deep space, beyond the global GPS system.

Machine translation

Online machine translation systems (e.g., Google Translate) now enable the reading of documents in over 100 languages, including the native languages of over 99% of humans, and render hundreds of billions of words per day for hundreds of millions of users.

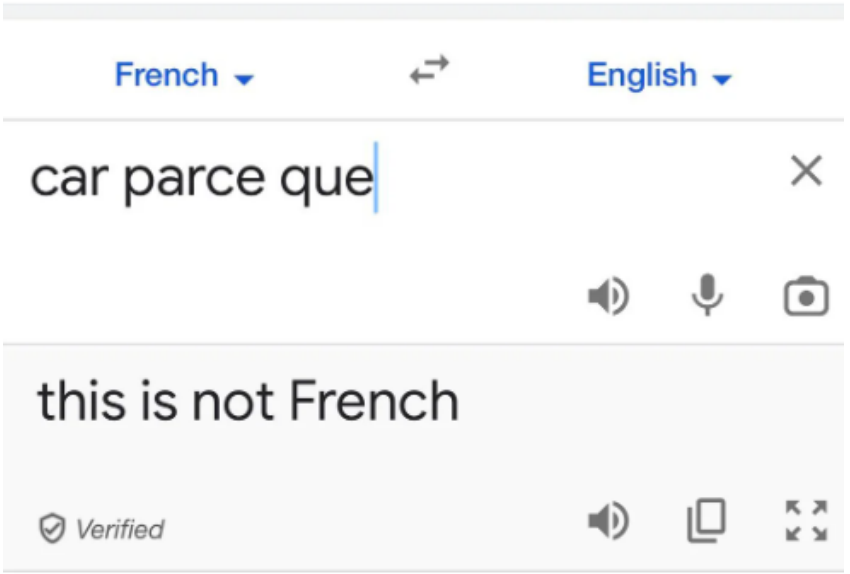


image source

Accuracy in translation of these systems is nowadays close to the level of a human.

Speech recognition and virtual assistants

In 2017, Microsoft showed that its Conversational Speech Recognition System had reached a word error rate of 5.1%, matching human performance on the Switchboard task, which involves transcribing telephone conversations.



image source

About a third of computer interaction worldwide is now done by voice rather than keyboard; Skype provides real-time speech-to-speech translation in ten languages. Alexa, Siri, Google offer virtual assistants that can answer questions and carry out tasks for the user.



image source

Recommendation

Companies such as Amazon, Facebook, Netflix, Spotify, YouTube, Walmart, and others use machine learning to recommend what you might like based on your past experiences and those of others like you.



image source

Spam filters



image source

Game playing

Deep Blue defeated world chess champion Garry Kasparov in 1997.

ALPHAGO surpassed all Go human players in 2017.

ALPHAGO benefited from studying hundreds of thousands of past games by human Go players, and from the distilled knowledge of expert Go players that worked on the team.

In 2018, a follow-up program, ALPHAZERO, used no input from humans (except for the rules of the game), and was able to learn through self-play alone to defeat all opponents, human and machine, at Go, chess, and shogi.

Human champions have been beaten by AI systems at games as diverse as Jeopardy! (2010), poker (2015; 2017; 2019), and the video games Dota 2 (2018), StarCraft II (2019), and Quake III (2019).



image source

Tools for predictions and decisions

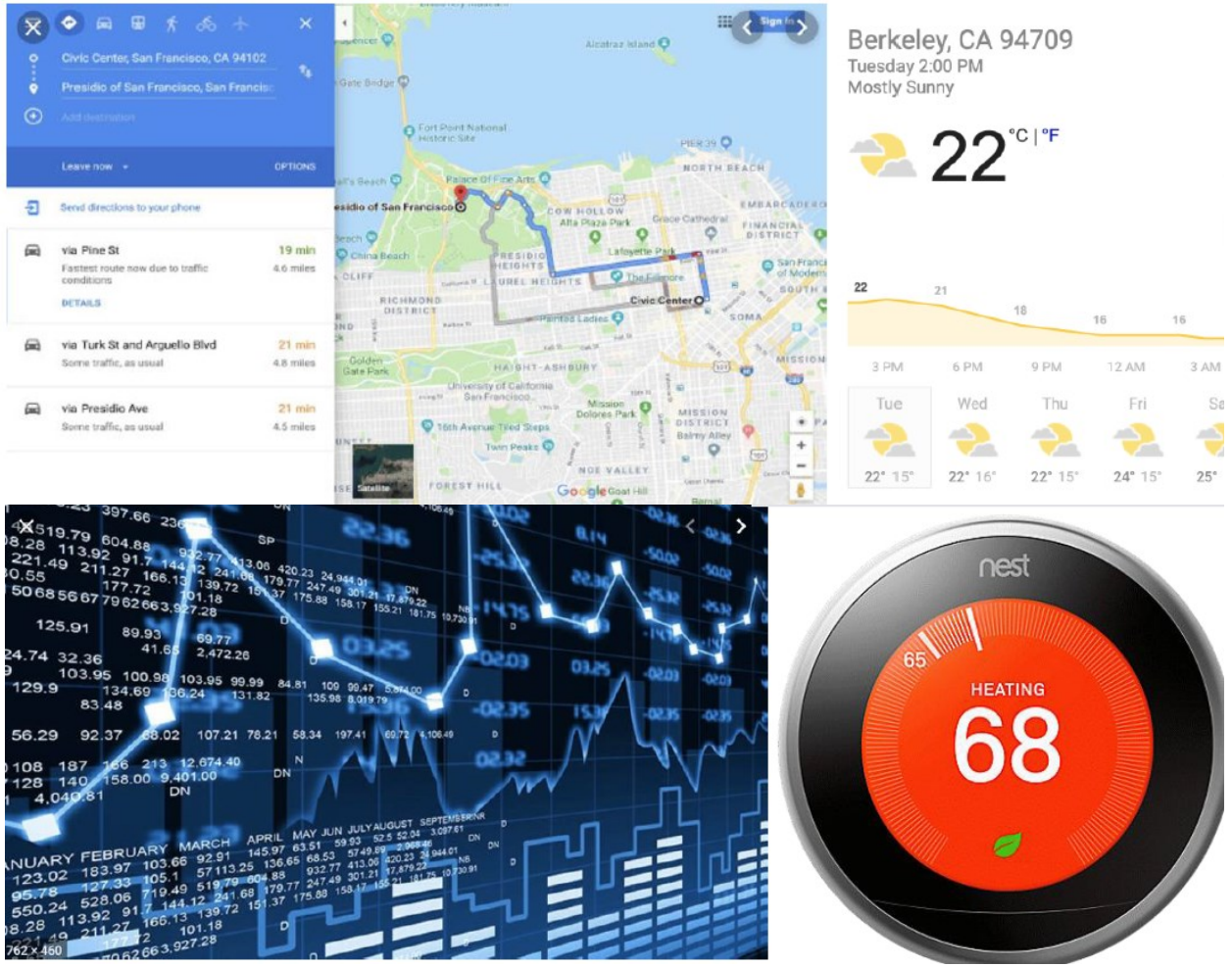


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Image understanding



"man in black shirt is playing guitar."



"construction worker in orange safety vest is working on road."



"two young girls are playing with lego toy."



"boy is doing backflip on wakeboard."



"girl in pink dress is jumping in air."



"black and white dog jumps over bar."



"young girl in pink shirt is swinging on swing."



"man in blue wetsuit is surfing on wave."

image source

Content generation: text, images, videos, art …

In 2022, an AI-generated work of art won the Colorado State Fair's art competition. The artist, Jason Allen, had used Midjourney – a generative AI system trained on art scraped from the internet – to create the piece. The process was far from fully automated: Allen went through some 900 iterations over 80 hours to create and refine his submission.

Yet his use of AI to win the art competition triggered a heated backlash online, with one Twitter user claiming, "We're watching the death of artistry unfold right before our eyes."



image source

Content generation: text, images, videos, art …



image source

DALL-E's view of Plato's account:

"*As Parmenides argued that movement could not exist, Zeno paced around.*"

Medicine

AI algorithms now equal or exceed expert doctors at diagnosing many conditions, particularly when the diagnosis is based on images.

For example, the LYNA system achieves 99.6% overall accuracy in diagnosing metastatic breast cancer—better than an unaided human expert—but the combination does better still (Liu et al., 2018; Steiner et al., 2018).

The widespread adoption of these techniques is now limited not by diagnostic accuracy but by the need to demonstrate improvement in clinical outcomes and to ensure transparency, lack of bias, and data privacy. In 2017, only two medical AI applications were approved by the FDA, but that increased to 12 in 2018, and continues to rise.

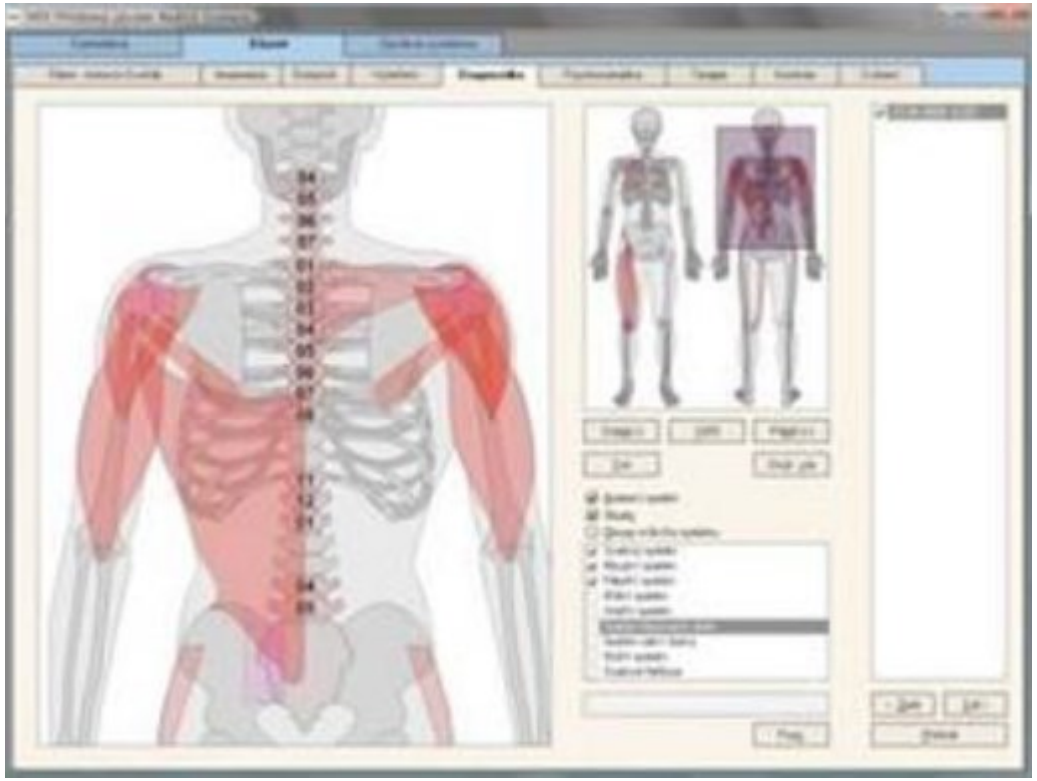


image source

Trading bots



image source

Climate science

A team of scientists from Berkeley Lab, Oak Ridge National Laboratory, and NVIDIA the 2018 ACM Gordon Bell Prize for a deep learning model that discovers detailed information about extreme weather events that were previously buried in climate data.

Rather than computing simple quantities such as average global temperature, the deep learning methods developed by the team discovered and located features like hurricanes and atmospheric rivers, which are important in characterizing extreme weather patterns and their impact.



image source

Chatbots

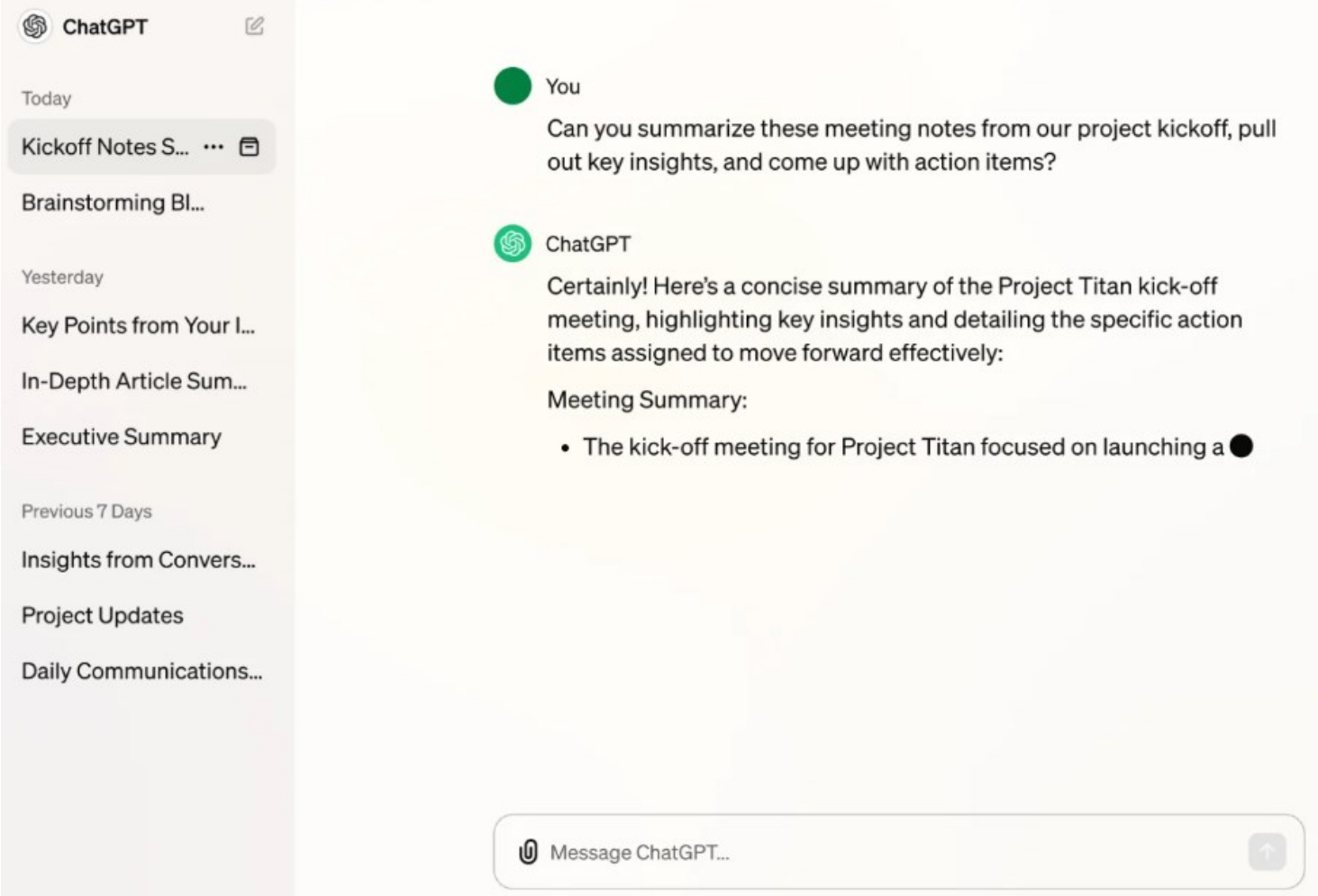


image source

OpenAI ChatGPT

Microsoft Copilot

Google Gemini

Meta AI

...

AI & job market

AI in industry

In the recent years, it has been observed an unprecedented proliferation of AI in industry.

AI is nowadays playing a vital role across a large variety of industries, revolutionizing the way how businesses work and providing new opportunities for

- innovation
- productivity improvement
- employment

AI market

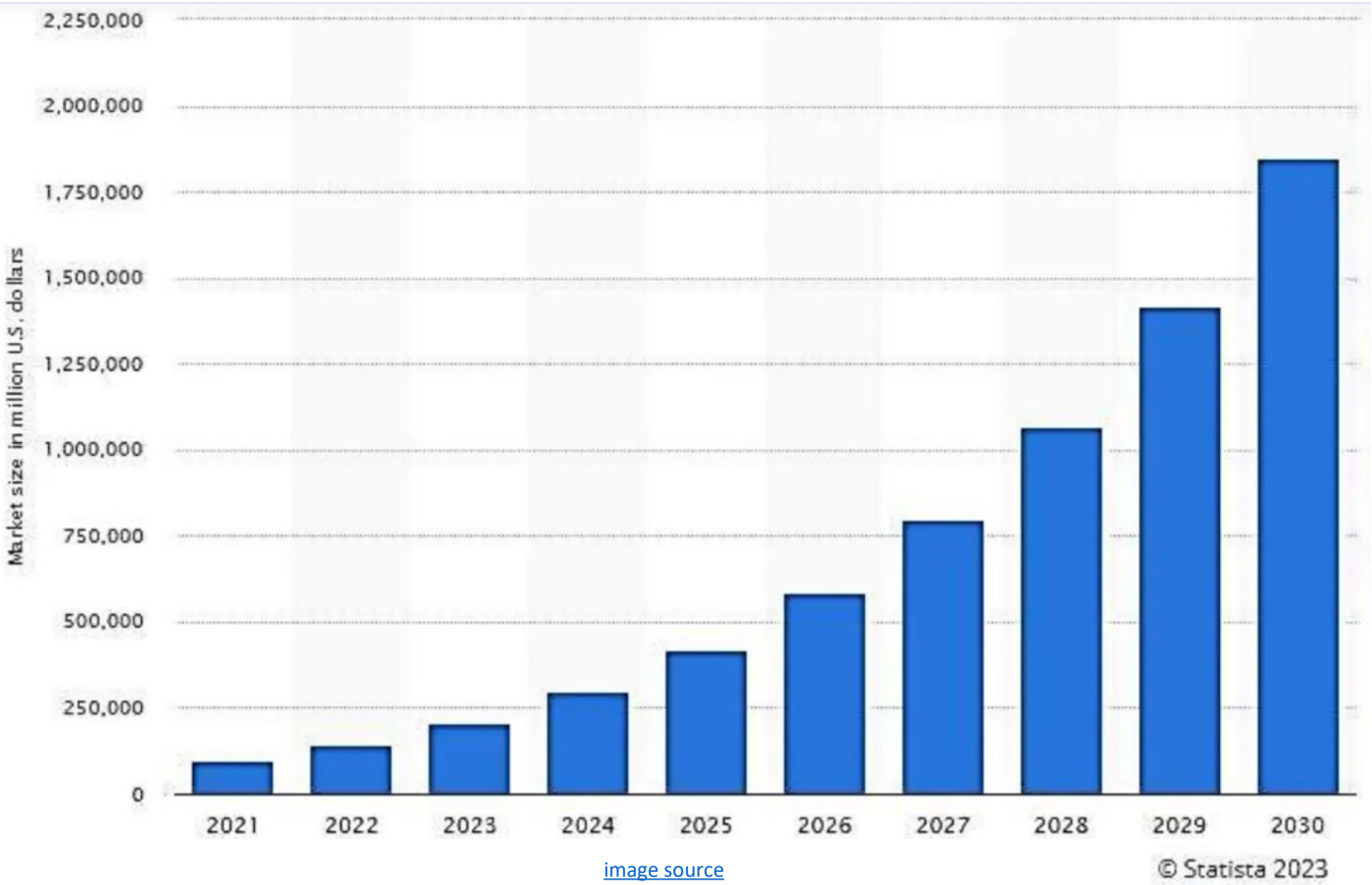


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- According to the Next Move Strategy Consulting, the AI sector will increase extremely rapidly over the next decade.
- Its worth is predicted to get bigger and bigger by 2030, reaching nearly 2 trillion US dollars.

AI market

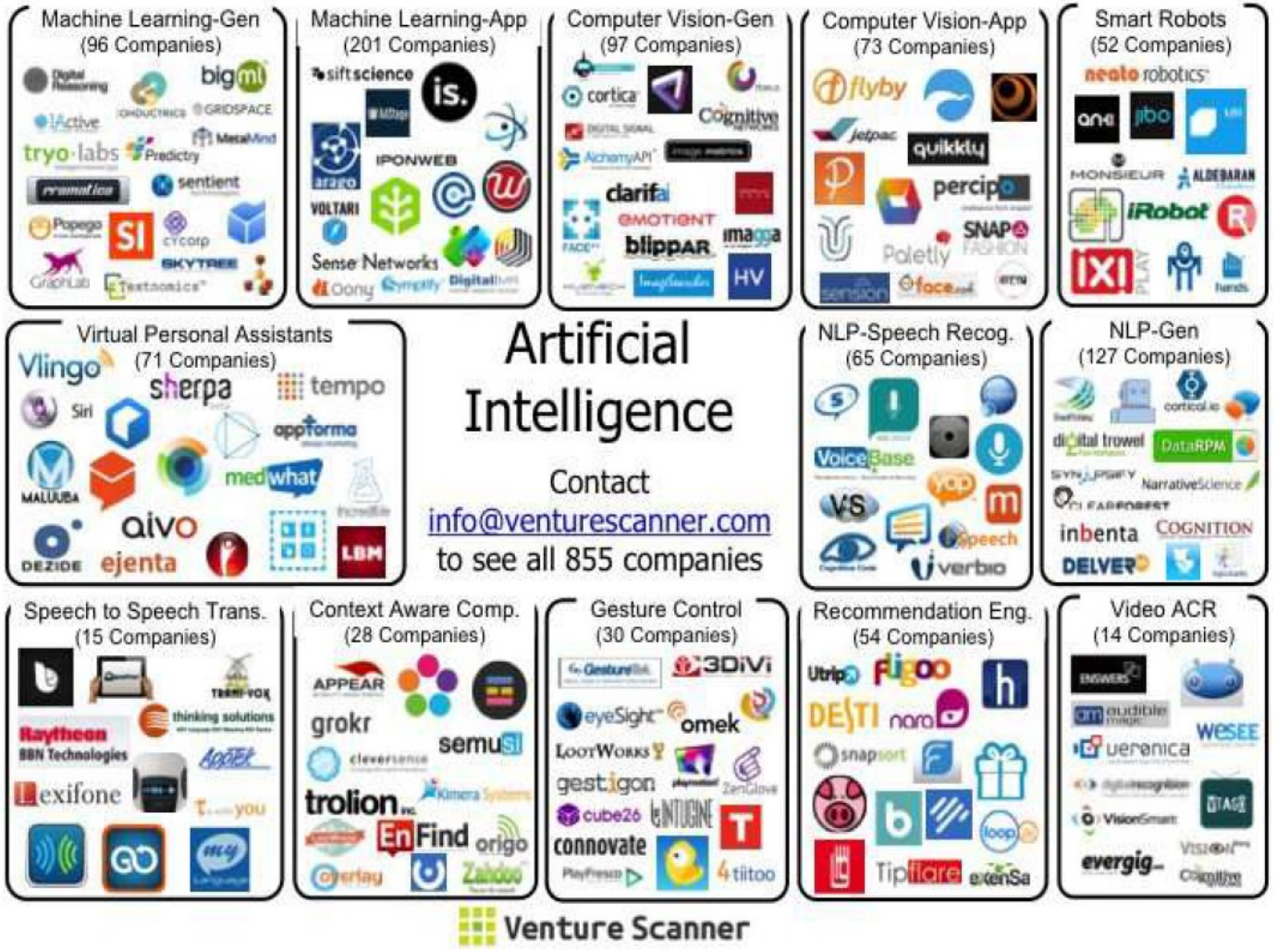


image source

AI employment opportunities

- AI has substantial potential to create significant employment opportunities across various sectors.
- AI fuels innovation and entrepreneurship across a wide range of industries.
- In coming slides we will discuss some examples of AI-driven industries that can create employment.



source

AI employment opportunities



Product and Service Enhancement

Applications

- AI-powered chatbots
- AI-driven analytics

Job Opportunities

- AI Customer Support Agents
- Chatbot Trainers
- Customer Experience Analysts



Healthcare and Life Sciences

Applications

- AI-powered Medical Devices
- Diagnostic Tools
- Health Monitoring Systems
- Personalized Treatment Plans
- Drug Discovery
- Medical Image Analysis

Job Opportunities

- AI Specialists
- Medical Data Analysts
- Healthcare AI Trainers
- Telemedicine Professionals



E-commerce and Retail

Applications

- Optimize inventory management and pricing strategies
- More relevant product suggestion

Job Opportunities

- AI-driven Marketing Specialists
- E-commerce Data Analyst
- Chatbot Developers
- Logistics Optimization Managers

source

AI employment opportunities



Financial Technology (FinTech)

Applications
- Fraud Detection
- Algorithmic Trading
- Credit Scoring
- Credit Risk Assessment
- Customer Support

Job Opportunities
- AI Financial Analysts
- Fintech Software Developers
- AI-driven Investment Advisors
- Blockchain Experts



Smart Manufacturing

Applications
- Smart Factories
- AI-driven Automation
- Predictive Maintenance

Job Opportunities
- Robotics Engineers
- Industrial AI Programmers
- Automation System Maintenance Technicians



Education and Training

Applications
- Personalized Learning
- Adaptive Tutoring
- Educational Content Creation

Job Opportunities
- AI Educators
- Instructional Designers
- Edtech Software Developers
- Data Analysts

source

AI employment opportunities



Smart Cities

Applications
- Traffic Management
- Waste Management
- Public Safety
- Energy Usage Optimization

Job Opportunities
- Urban AI Planners
- Smart City Project Managers
- IoT Integration Specialists



Entertainment and Media

Applications
- Content Recommendation
- Personalized Content Creation
- AI-generated Art, Music, & Content
- Immersive Experiences

Job Opportunities
- AI Content Creators
- VR/AR Developers
- Media Data Analysts



Agricultural Innovation

Applications
- AI-powered sensors and drones
- Precision Agriculture
- Crop Management Optimization
- Resource Utilization
- Yield Prediction

Job Opportunities
- Agricultural Data Analysts
- Precision Agriculture Consultants
- AI Agricultural Engineers

source

AI employment opportunities



Transportation and Mobility

Applications
- AI-driven Transportation Solutions
- Innovations in Autonomous Vehicles
- Traffic Management

Job Opportunities
- AI Vehicle Engineers
- Transportation System Planners
- Autonomous Vehicle Safety Testers



Energy and Utilities

Applications
- AI-driven Energy Management
- Grid Optimization

Job Opportunities
- Energy Data Analysts
- AI Energy Consultants
- Smart Grid Engineers



Environmental Monitoring and Conservation

Applications
- Analyze Environmental Data For Conservation Efforts

Job Opportunities
- Environmental Data Scientists
- AI Ecologists
- Conservation Technology Specialists

source

AI employment opportunities



Transportation and Mobility

Applications
- AI-driven Transportation Solutions
- Innovations in Autonomous Vehicles
- Traffic Management

Job Opportunities
- AI Vehicle Engineers
- Transportation System Planners
- Autonomous Vehicle Safety Testers



Energy and Utilities

Applications
- AI-driven Energy Management
- Grid Optimization

Job Opportunities
- Energy Data Analysts
- AI Energy Consultants
- Smart Grid Engineers



Environmental Monitoring and Conservation

Applications
- Analyze Environmental Data For Conservation Efforts

Job Opportunities
- Environmental Data Scientists
- AI Ecologists
- Conservation Technology Specialists



Cybersecurity

Applications
- Threat Detection
- Anomaly Detection
- Real-time Cybersecurity Monitoring.

Job Opportunities
- AI Cybersecurity Analysts
- Penetration Testers
- AI-driven Security Software Developers

source

AI-driven job market



source

- AI/machine-learning (ML) engineer
- AI research scientist
- AI architect
- AI Quality Assurance (QA) engineer
- AI product manager
- Data Scientist
- AI/ML ethics officer
- Natural Language Processing (NLP) engineer
- Computer Vision engineer
- AI trainer/annotation specialist

What AI is

AI systems

An AI (computing) system is a system capable of mimicking intelligent behavior that is typical of human beings.

Capabilities of an AI system may include:

- learning; reasoning; perceiving; acting; experiencing sentience, emotions, or consciousness

Prominent examples of AI systems:

- Virtual assistants (e.g., Siri, Alexa, Google Assistant)
- Recommender systems (e.g., the ones that recommend movies on Netflix, or products on Amazon)
- Chatbots (e.g., ChatGPT)
- Self-driving cars
- Facial-recognition software
- Robotic vacuum cleaners
- Systems/software capable of: playing chess, proving mathematical theorems, writing poetry, generating art

AI systems: definitions

There are two main schools of thought for defining an AI system:

1. A system exhibiting intelligence as a property of internal thought processes and reasoning,
   - Focus on the methods that are used to build an AI system (the "how")
2. A system exhibiting intelligent behavior, as an external characterization.
   - Focus on the objective specification of what we want an AI system to do (the "what")

Moreover, intelligence can be considered:

- Either in terms of fidelity to human performance,
- Or, as an abstract, formal definition, called rationality, i.e., loosely speaking, doing the "right thing".
  - Rationality differs from human performance not because humans are meant to be "irrational" in the dictionary sense of "deprived of normal mental clarity". Rather, here it is conceded that human decisions are not always perfect (unlike rationality), according to some mathematical definition of "perfection".

AI systems: definitions

Therefore, there are two (orthogonal) dimensions along which an AI system can be defined:

- Humanity vs. rationality
- Thought vs. behavior

There are 4 possible combinations of these factors; there have been adherents and research programs for all 4 such combinations:

- Acting Humanly: AI that mimics human behavior, such as the Turing Test (see next).
- Thinking Humanly: AI that replicates human thought processes, focusing on cognitive modeling.
- Thinking Rationally: AI that that makes rational decisions, using logic and reasoning.
- Acting Rationally: AI that takes actions to achieve the best outcomes.

Acting humanly: the Turing test

The Turing test was designed as a thought experiment that would sidestep the philosophical vagueness of the question "Can a machine think?".

Designed by Alan Turing, the famous English scientist who originally provided the foundations of computation, and was widely considered the father of (theoretical) computer science.

Turing designed the Turing test in his famous 1950's paper titled "*Computing Machinery and Intelligence*". In that paper, the Turing test was originally called "imitation game".



Turing in 1936

image source

The Turing test

The Turing test contains three terminals, which are separated from one another physically. Two terminals are occupied by humans, whereas the third is occupied by a computer.

One of the humans serves as a questioner, whereas the human and the computer at the other two terminals are respondents.

Since the terminals are separated from one another physically, the human questioner does not know which of the two terminals is occupied by the respondent human and which terminal is occupied by the computer that is being tested for its artificial intelligence abilities. Similarly, the human and the computer respondent do not have access to each other's responses.

The questioner poses the same question to the human and the computer to test how "human-like" their responses might be. If the human questioner is unable to tell the difference between the human respondent and the computer respondent, the computer respondent is said to have passed the Turing test.

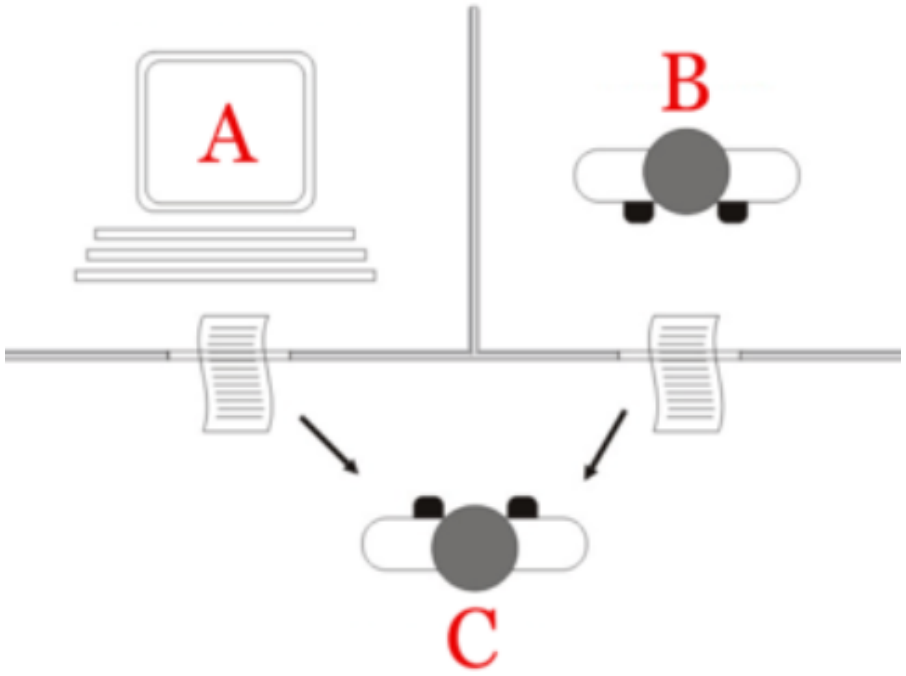


image source

The Turing test: required capabilities

To pass the Turing test, a computer would need at least the following capabilities:

- natural language processing: understanding and generating human language
  - to communicate successfully in a human language;
- knowledge representation: storing and utilizing knowledge about the world
  - to store what it knows or hears;
- automated reasoning: drawing logical conclusions and solving problems
  - to answer questions and to draw new conclusions;
- machine learning: learning from examples
  - to detect and extrapolate patterns, to be used for adapting to new circumstances.

The total Turing test

Turing viewed the physical simulation of a person as unnecessary to demonstrate intelligence.

However, other researchers have proposed a total Turing test, which requires interaction with objects and people in the real world. To pass the total Turing test, a robot will need:

- computer vision (acquiring, processing, analyzing and understanding images, videos and other visual inputs) and speech recognition (translation of human speech from a verbal format to a text one) to perceive the world;
- robotics (design, construction, operation, and use of robots) to manipulate objects and move about.

Limitations of Turing test

- Not reproducible or amenable to mathematical analysis.
- It does not rely on a rigorous definition of intelligence.
- AI researchers devoted little effort to make systems pass the Turing test.
- It implicitly assume analogy between "behaving intelligently" and "behaving humanly", which somehow implies that emulating humans is the right thing to do to achieve AI.
  - Shouldn't we study the underlying principles of intelligence instead? Successful flight machines have not been developed by imitating birds, rather by studying engines and aerodynamics.

The history of AI

The inception of artificial intelligence (1943–1956)

The first work that is nowadays generally recognized as AI was done by Warren McCulloch and Walter Pitts in 1943.

- They devised the first model of artificial neuron, characterized as being “on” or “off,” with a switch to “on” occurring in response to stimulation by a sufficient number of neighboring neurons. The state of a neuron was conceived of as “factually equivalent to a proposition which proposed its adequate stimulus”.



image source

The inception of artificial intelligence (1943–1956)

Donald Hebb (1949) demonstrated a simple updating rule for modifying the connection strengths between neurons. His rule, now called Hebbian learning, was used in many early neural networks, such as Rosenblatt's perceptron and the Hopfield network.

Two undergraduate students at Harvard, Marvin Minsky and Dean Edmonds, built the first neural network computer in 1950. The "SNARC" system, as it was called, used 3000 vacuum tubes and a surplus automatic pilot mechanism from a B-24 bomber to simulate a network of 40 neurons.

The inception of artificial intelligence (1943–1956)

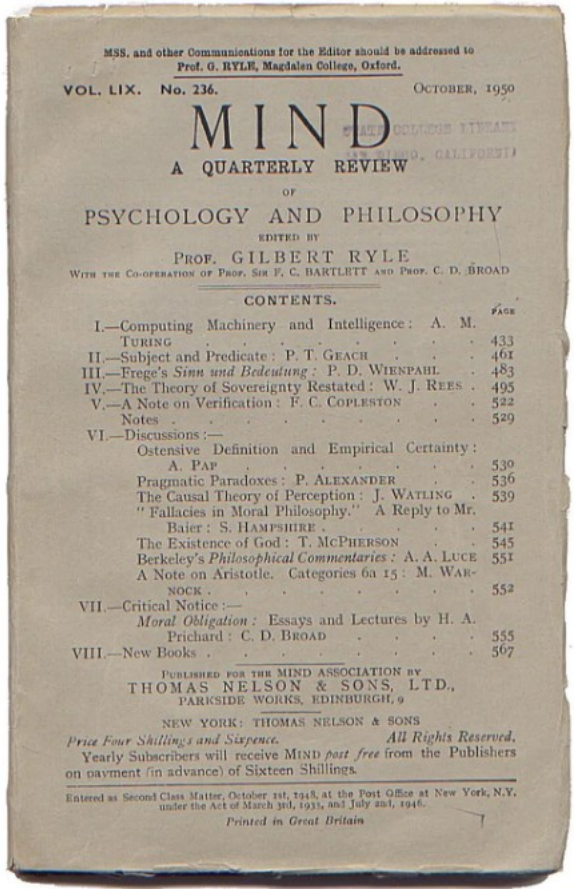
Alan Turing's vision about AI started as early as 1947, with a series of lectures on the topic at the London Mathematical Society and culminated into a persuasive agenda in his seminal 1950's paper titled "Computing Machinery and Intelligence". Therein, he:

- introduced the Turing test, machine learning, genetic algorithms, and reinforcement learning.
- dealt with many of the objections raised to the possibility of AI that have been arisen since then
- suggested that it would be easier to create human-level AI by developing learning algorithms and then teaching the machine rather than by programming its intelligence by hand
- warned that achieving AI goals might not be the best thing for the human race



Turing in 1936

image source



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MIND

A QUARTERLY REVIEW

OF

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The inception of artificial intelligence (1943–1956)

Birth of AI: the Dartmouth workshop in 1956



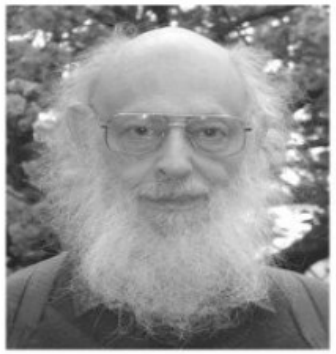
John MacCarthy



Marvin Minsky



Claude Shannon



Ray Solomonoff



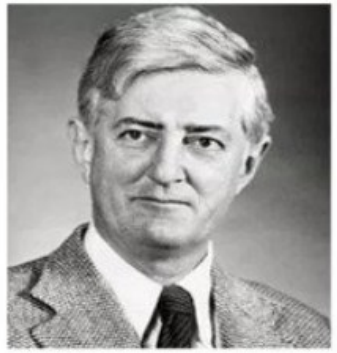
Alan Newell



Herbert Simon



Arthur Samuel



Oliver Selfridge



Nathaniel Rochester



Trenchard More

image source

The inception of artificial intelligence (1943–1956)

The 2-month workshop organized by John McCarthy of Dartmouth College, held at Dartmouth during the summer of 1956, is generally considered the birthplace of AI as a formal field, where the term ‘Artificial Intelligence’ was coined and used for the first time.

The workshop set the agenda for AI research, focusing on topics like reasoning, language understanding, and learning.

- Main takeaway of the workshop: every aspect of learning or any other feature of intelligence can be so precisely described that a machine can be made to simulate it.



image source

At the Dartmouth workshop (1956), Allen Newell and Herbert Simon from Carnegie Tech presented a mathematical theorem-proving system called the Logic Theorist (LT). The program was able to prove most of the theorems in Chapter 2 of Russell and Whitehead’s “Principia Mathematica”. Russell was reportedly delighted when told that LT had come up with a proof for one theorem that was shorter than the one in Principia.

Early enthusiasm, great expectations (1952–1969)

The intellectual establishment of the 1950s, by and large, preferred to believe that "a machine can never do X."

AI researchers naturally responded by demonstrating one X after another. They focused in particular on tasks considered indicative of intelligence in humans, including games, puzzles, mathematics, and IQ tests.

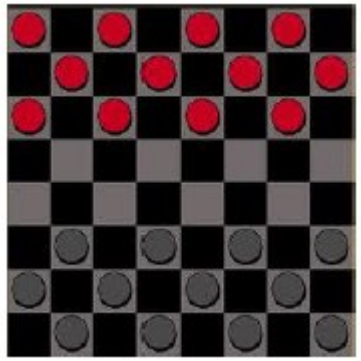


image source

Arthur Samuel (1956) devised a program to play checkers (draughts) which employed methods that we now call reinforcement learning. Samuel's programs learned to play at a strong amateur level.

Early enthusiasm, great expectations (1952–1969)

Newell, Shaw, and Simon (1957) followed up their success with LT with the General Problem Solver, or GPS. Unlike LT, this program was designed from the start to imitate human problem solving protocols. GPS was probably the first program to embody the "thinking humanly" approach.

The success of GPS and subsequent programs as models of cognition led Newell and Simon (1976) to formulate the famous physical symbol system hypothesis, which states that "a physical symbol system has the necessary and sufficient means for general intelligent action". What they meant is that any system (human or machine) exhibiting intelligence must operate by manipulating data structures composed of symbols.

- This hypothesis has been challenged from many directions, see later in the course

Early enthusiasm, great expectations (1952–1969)

In 1958, John McCarthy made two important contributions to AI:

1. He defined the high-level programming language Lisp, which was to become the dominant AI programming language for the next 30 years.
2. In a paper entitled “Programs with Common Sense”, he presented the Advice Taker, a hypothetical program that would embody general knowledge of the world and could use it to derive plans of action based on logic.
   - The Advice Taker thus embodied the central principles of knowledge representation and reasoning: have a formal, explicit representation of the world and its workings and to be able to manipulate that representation with deductive processes in order to perform reasoning.
   - The paper strongly influenced the course of AI and remains relevant today.



image source

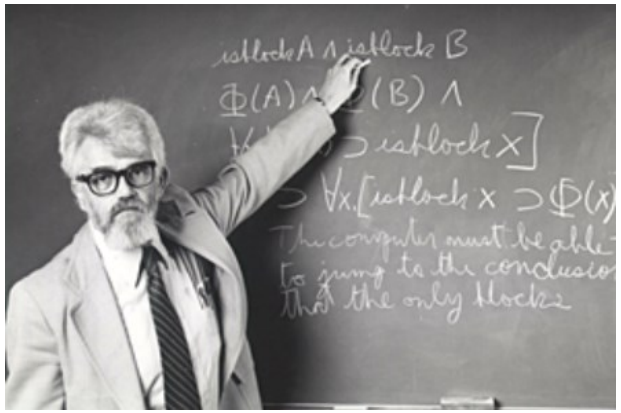


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Early enthusiasm, great expectations (1952–1969)

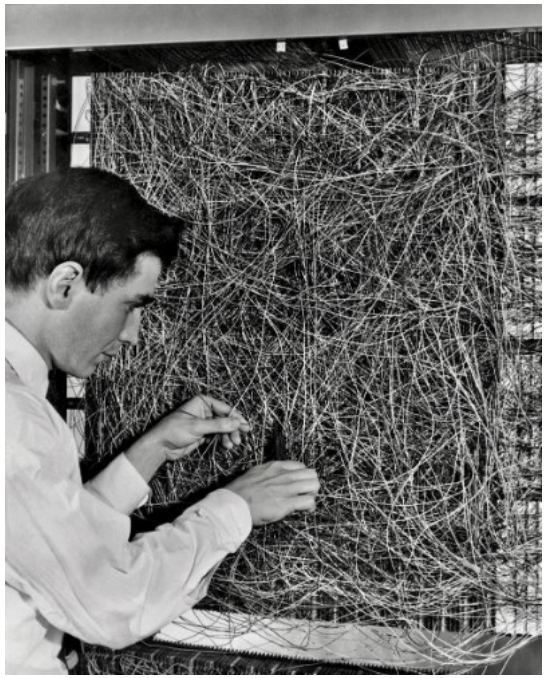
In 1958, psychologist Frank Rosenblatt invented the perceptron, the first implemented artificial neural network, funded by the United States Office of Naval Research.

At that time, computers were not very flexible, so the model is very simple and was implemented in hardware.

It was inspired by a simple model of a neuron, and was originally designed for image recognition.

The invention of the perceptron raised public excitement for research in artificial neural networks, causing the US government to drastically increase funding into deep learning research. This led to "the Golden Age of AI" fueled by the optimistic claims made by computer scientists regarding the ability of perceptrons to emulate human intelligence.

image source



Frank Rosenblatt working on the Mark I Perceptron computer in 1960

A dose of reality (1966–1973)

Herbert Simon in 1957 predicted that within 10 years a computer would be chess champion and a significant mathematical theorem would be proved by machine. These predictions came true (or approximately true) within 40 years rather than 10.

Simon's overconfidence was due to the promising performance of early AI systems on simple examples. In almost all cases, however, these early systems failed on more difficult problems.

A dose of reality (1966–1973)

Main reasons of failure in Simon's prediction:

1. Early AI systems were based primarily on "informed introspection" as to how humans perform a task, rather than on a careful analysis of the task, what it means to be a solution, and what an algorithm would need to do to reliably produce such solutions.
2. Intractability of many of the problems that AI was attempting to solve. Most of the early problem-solving systems worked by trying out different combinations of steps until the solution was found. This strategy worked initially because only small-sized problem instances were considered.
    - Before the theory of computational complexity was developed, it was widely thought that "scaling up" to larger problems was simply a matter of faster hardware and larger memories.

A dose of reality (1966–1973)

The illusion of unlimited computational power was not confined to problem-solving programs.

Early experiments in machine evolution (now called genetic programming) (Friedberg, 1958; Friedberg et al., 1959) were based on the undoubtedly correct belief that by making an appropriate series of small mutations to a machine-code program, one can generate a program with good performance for any particular task. The idea, then, was to try random mutations with a selection process to preserve mutations that seemed useful.

Despite thousands of hours of CPU time, almost no progress was demonstrated.

A dose of reality (1966–1973)

Minsky and Papert's book "Perceptrons" (1969) proved that, although perceptrons (introduced by Rosenblatt in 1958) could be shown to learn anything they were capable of representing, they could represent very little.

In particular, a two-input perceptron could not be trained to recognize when its two inputs were different (XOR function). Although their results did not apply to more complex, multilayer networks, this was enough to make research funding for neural-network research soon dwindled to almost nothing.

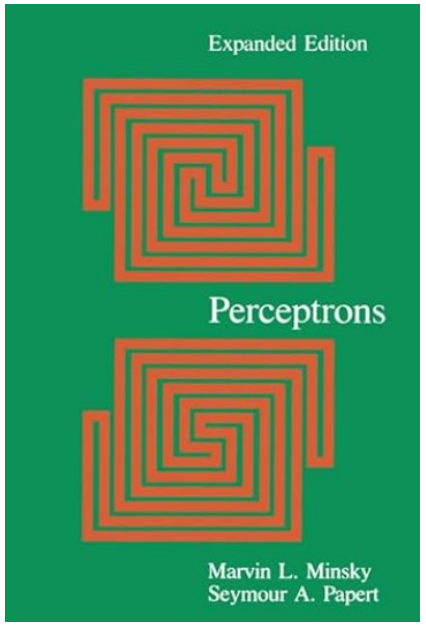


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Minsky and Papert's book is the center of a long-standing controversy in the study of artificial intelligence.

It is generally acknowledged that the pessimistic results provided by the authors in the book were responsible for the little research conducted on neural networks in the 1970s and 1980s, giving rise to what is called a "neural network winter".

During such a neural network winter, a change in the direction of research in AI was made, concentrating efforts on so-called "symbolic" systems.

Expert systems (1969–1986)

The picture of problem solving that had arisen during the first decade of AI research was of a general-purpose search mechanism trying to string together elementary reasoning steps to find complete solutions. Such approaches have been called weak methods because, although general, they do not scale up to large or difficult problem instances.

The alternative to weak methods is to use more powerful, domain-specific knowledge that allows larger reasoning steps and can more easily handle typically occurring cases in narrow areas of expertise.

Expert systems (1969–1986)

The DENDRAL program (Buchanan et al., 1969) was an early example of this approach.

It was developed at Stanford, where Ed Feigenbaum (a former student of Herbert Simon), Bruce Buchanan (a philosopher turned computer scientist), and Joshua Lederberg (a Nobel laureate geneticist) teamed up to solve the problem of inferring molecular structure from the information provided by a mass spectrometer.

The relevance of DENDRAL was that it was the first successful knowledge-intensive system: its expertise derived from large numbers of special-purpose rules.

In 1971, Feigenbaum and others at Stanford began the Heuristic Programming Project (HPP) to investigate the extent to which the new methodology of expert systems could be applied to other areas.

Expert systems (1969–1986) and the “AI winter”

DENDRAL was the pioneer of many subsequent expert systems focusing on applications to real-world problems.

This has led to the development of a wide range of representation and reasoning tools. Some were based on logic: for example, the Prolog language became popular in Europe and Japan, and the PLANNER family in the United States.

Overall, the AI industry boomed from a few million dollars in 1980 to billions of dollars in 1988, including hundreds of companies building expert systems, vision systems, robots, and software and hardware specialized for these purposes.

Soon after that came a period called the “AI winter,” in which many companies fell by the wayside as they failed to deliver on extravagant promises. It turned out to be difficult to build and maintain expert systems for complex domains, in part because the reasoning methods used by the systems broke down in the face of uncertainty and in part because the systems could not learn from experience.



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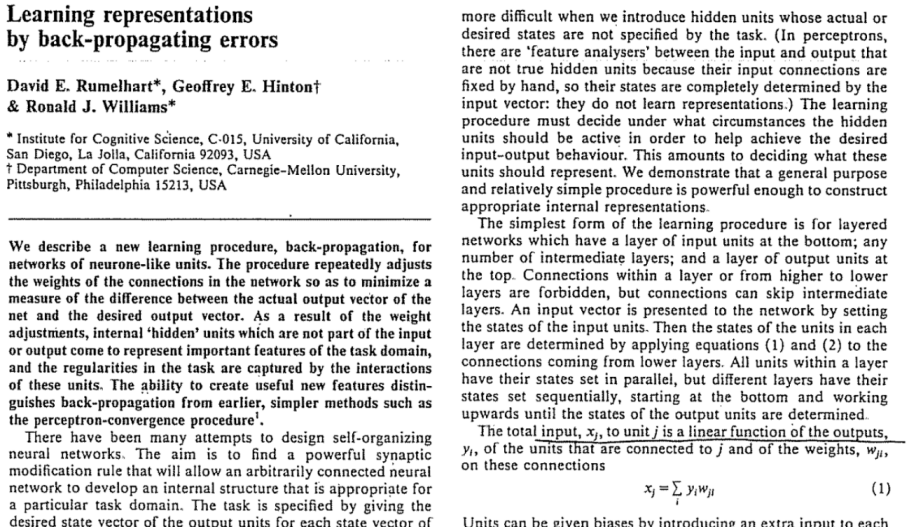
The return of neural networks (1986–present)

A resurge of neural network arose in the mid-1980s, with the (re)invention of the popular backpropagation algorithm, which was the key to efficiently training neural networks composed of multilayer perceptrons.

The actual invention of the backpropagation algorithm roots back to the early 1960s, even though in different contexts (Kelley, 1960; Bryson, 1962).

The reinvention of backpropagation in the mid-1980s was independently delivered by at least four different groups of researchers.

But, it is common to attribute its actual reinvention to the work of Rumelhart, Hinton, and Williams, culminated in the famous 1986's Nature paper titled "Learning representations by back-propagating errors".



Learning representations by back-propagating errors

David E. Rumelhart*, Geoffrey E. Hinton† & Ronald J. Williams*

* Institute for Cognitive Science, C-015, University of California, San Diego, La Jolla, California 92093, USA
† Department of Computer Science, Carnegie-Mellon University, Pittsburgh, Philadelphia 15213, USA

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure[1].

There have been many attempts to design self-organizing neural networks. The aim is to find a powerful synaptic modification rule that will allow an arbitrarily connected neural network to develop an internal structure that is appropriate for a particular task domain. The task is specified by giving the desired state vector of the output units for each state vector of

more difficult when we introduce hidden units whose actual or desired states are not specified by the task. (In perceptrons, there are 'feature analysers' between the input and output that are not true hidden units because their input connections are fixed by hand, so their states are completely determined by the input vector: they do not learn representations.) The learning procedure must decide under what circumstances the hidden units should be active in order to help achieve the desired input–output behaviour. This amounts to deciding what these units should represent. We demonstrate that a general purpose and relatively simple procedure is powerful enough to construct appropriate internal representations.

The simplest form of the learning procedure is for layered networks which have a layer of input units at the bottom; any number of intermediate layers; and a layer of output units at the top. Connections within a layer or from higher to lower layers are forbidden, but connections can skip intermediate layers. An input vector is presented to the network by setting the states of the input units. Then the states of the units in each layer are determined by applying equations (1) and (2) to the connections coming from lower layers. All units within a layer have their states set in parallel, but different layers have their states set sequentially, starting at the bottom and working upwards until the states of the output units are determined.

The total input, x_j, to unit j is a linear function of the outputs, y_i, of the units that are connected to j and of the weights, w_{ji}, on these connections

$$x_j = \sum_i y_i w_{ji} \quad (1)$$

Units can be given biases by introducing an extra input to each

image source

The introduction of this algorithm rekindled an interest in neural networks. However, several computational, stability, and overfitting challenges were found in the use of this algorithm. As a result, research in the field of neural networks again fell from favor.

Probabilistic reasoning and machine learning (1987–present)

The brittleness of expert systems led to a new, more scientific approach incorporating probability rather than Boolean logic, machine learning rather than hand-coding, and experimental results rather than philosophical claims.

Judea Pearl's (1988) "Probabilistic Reasoning in Intelligent Systems" led to a new acceptance of probability and decision theory in AI. Pearl's development of Bayesian networks yielded a rigorous and efficient Bayesian network formalism for representing uncertain knowledge as well as practical algorithms for probabilistic reasoning.

A second major contribution in 1988 was Rich Sutton's work connecting reinforcement learning—which had been used in Arthur Samuel's checker-playing program in the 1950s—to the theory of Markov decision processes (MDPs) developed in the field of operations research.

A flood of work followed connecting AI planning research to MDPs, and the field of reinforcement learning found applications in robotics and process control as well as acquiring deep theoretical foundations.

Big data (2001–present)

Remarkable advances in computing power and the creation of the World Wide Web have facilitated the creation of very large data sets—a phenomenon sometimes known as big data.

These datasets may include trillions of words of text, billions of images, and billions of hours of speech and video, as well as vast amounts of genomic data, vehicle tracking data, clickstream data, social network data, and so on.

This has led to the development of learning algorithms specially designed to take advantage of very large datasets.

Banko and Brill from Microsoft Research, in their 2001's paper about natural language disambiguation, argued that the improvement in performance obtained from increasing the size of the data set by two or three orders of magnitude outweighs any improvement that can be obtained from tweaking the algorithm.

The availability of big data and the shift towards machine learning helped AI recover commercial attractiveness (Havenstein, 2005; Halevy et al., 2009). Big data was a crucial factor in the 2011 victory of IBM's Watson system over human champions in the Jeopardy! Quiz game, an event that had a major impact on the public's perception of AI.



image source

Deep learning (2011–present)

The term deep learning refers to machine learning using many layers of simple, adjustable computing elements.

In the context of neural networks, this corresponds to having many layers of perceptrons.

Experiments were carried out with such networks as far back as the 1970s, and in the form of convolutional neural networks they found some success in handwritten digit recognition in the 1990s (LeCun et al., 1995).

It was not until 2011, however, that deep learning methods really took off, mainly thanks to the development of more powerful and dedicated hardware (e.g., GPU, TPU).

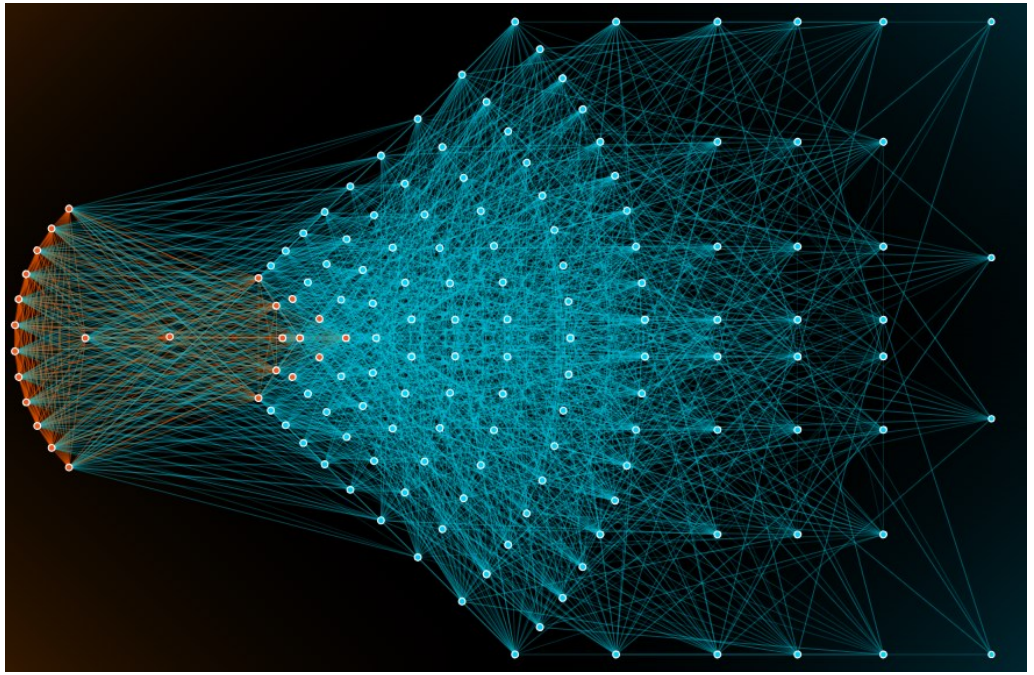


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Deep learning (2011–present)

In the 2012 ImageNet competition, which required classifying images into one of a thousand categories (armadillo, bookshelf, corkscrew, etc.), a deep learning system created in Geoffrey Hinton's group at the University of Toronto (Krizhevsky et al., 2013) demonstrated a dramatic improvement over previous systems, which were based largely on handcrafted features.

Since then, deep learning systems have exceeded human performance on some vision tasks (and lag behind in some other tasks). Similar gains have been reported in speech recognition, machine translation, medical diagnosis, and game playing.

The use of a deep network to represent the evaluation function contributed to ALPHAGO's victories (in 2016-2018) over the leading human Go players.

Deep learning (2011–present)

These remarkable successes have led to a resurgence of interest in AI among students, companies, investors, governments, the media, and the general public.

In 2014, the state of the art was training “very deep neural network” with 20 to 30 layers.

Successfully training such deep networks also depends on the availability of large amounts of training data, and on a few algorithmic tricks.

Deep learning (2011–present)

image source



ACM named Yoshua Bengio, Geoffrey Hinton, and Yann LeCun recipients of the 2018 ACM A.M. Turing Award for conceptual and engineering breakthroughs that have made deep neural networks a critical component of computing.

Deep learning (2011–present)

A key milestone in deep learning was placed in 2017, when Ashish Vaswani et al. introduced the modern Transformer architecture in their seminal paper "Attention Is All You Need".

Transformers have revolutionized the deep learning landscape.

Among others, Transformers have increasingly become the model of choice for natural language processing. Many modern large language models such as GPT-x or LLaMA, use the Transformer architecture.

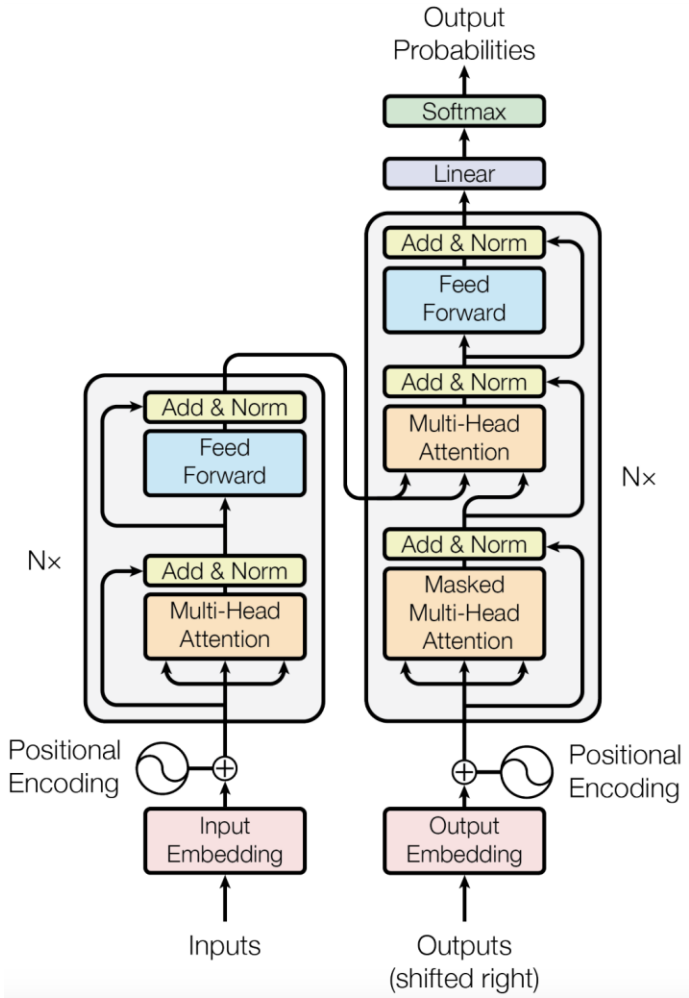


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AI boom (late 2010s-present)

The AI boom, or AI spring, is an ongoing period of rapid progress in the field of artificial intelligence (AI) that started in the late 2010s before gaining international prominence in the early 2020s.



image source

Prominent results include protein folding prediction led by Google DeepMind (with their AlphaFold system) and generative AI systems (e.g., ChatGPT) developed by OpenAI.

Forms of AI:
Artificial General Intelligence vs. Narrow AI

Artificial General Intelligence vs. Narrow AI

Most modern forms of artificial intelligence are application- and domain-specific, where the system is taught to perform a relatively narrow task (e.g., generate text or visual content, play chess, categorize an image, etc.).

- This type of AI is often referred to as "Narrow AI"



image source

However, the original goal of artificial intelligence was to develop general-purpose intelligence.

- The subfield of AI that deals with the development of general forms of intelligence is referred to as Artificial General Intelligence (AGI).

Strong AI vs. weak AI

AGI is also known as:

- strong AI (even though this term is often reserved for systems that experience sentience or consciousness)
- full AI
- human-level AI

Narrow AI is instead also referred to as “weak AI”.

However, some academic sources use "weak AI" to refer more broadly to any system that neither experience consciousness nor have a mind in the same sense as humans.

Artificial General Intelligence: examples

Perhaps the earliest prominent example of a system aspiring to achieve the AGI goal was the GPS (General Problem Solver) general-purpose problem solver system created by Newell, Shaw, and Simon in 1957 (overviewed a few slides back).

Another more recent example of an AGI system is Cyc (where the term "cyc" comes from "encyclopedia"), which is a large-scale project that has been in progress since the 1980s, whose goal is to assemble a comprehensive ontology and knowledge base that spans the basic concepts and rules about how the world works.

- While this system does have some commercial applications in specific domains, its utility as general-purpose reasoning system is quite rudimentary (and it fails to achieve the reasoning capabilities of a small child).



image source

How far we are from AGI

So far, no AI system has satisfactorily achieved the goal of AGI.

This has led to the natural question as to what the ultimate goal of AGI might be.

Also, the timeline for achieving AGI remains a subject of ongoing debate among researchers and experts.

As of 2023, some argue that it may be possible in years or decades; others maintain it might take a century or longer; and a minority believe it may never be achieved (https://ourworldindata.org/ai-timelines).

AGI and modern LLMs

There is debate on the exact definition of AGI, and regarding whether modern large language models (LLMs) such as GPT-4 – the LLM behind the popular ChatGPT tool – are early, incomplete forms of AGI (https://futurism.com/gpt-4-sparks-of-agi).

A framework for classifying AGI in levels was proposed in 2023 by Google DeepMind researchers.

They define five levels of AGI: emerging, competent, expert, virtuoso, and superhuman. For example, a competent AGI is defined as an AI that outperforms 50% of skilled adults in a wide range of non-physical tasks, and a superhuman AGI is similarly defined but with a threshold of 100%.

They consider modern LLMs like GPT-4 or LLaMA 2 to be instances of emerging AGI (https://venturebeat.com/ai/here-is-how-far-we-are-to-achieving-agi-according-to-deepmind/).

Forms of AI: Symbolic AI

Symbolic AI

Symbolic AI encompasses all those methodologies for building AI systems which are based on manipulating formal symbolic representations of the underlying problems to be tackled and solved by the target AI system, and performing abstract reasoning over those symbols.

Like the symbols of mathematics or logic, symbols in symbolic AI – and AI systems based on them – can be regarded as purely formal (meaningless) structures.

In practice, however, they are normally interpreted by the user in terms of some particular semantic content: verbal, numerical, visual, auditory, and so on.

Often such symbols correspond to linguistic or logical expressions aimed at representing the knowledge of the underlying AI system.

As such, symbols of symbolic AI are actually human-readable and AI systems based on them are interpretable by design.

Atomic symbols can be combined to form complex symbols, according to specific formal rules of symbol manipulation. What happens when a symbolic AI system is run is that symbols of various kinds, structured in various ways, are built, stored, retrieved, compared, and transformed.

Symbolic AI: examples
Expert systems, decision support systems

Medical diagnosis systems typically represents the knowledge about the medical domain through symbols that conceptually correspond to notions like “patient”, “doctor”, “disease”, “symptom”, “diagnosis”, “cure”, and so on.

Diagnoses are made based on semantic rules that perform reasoning over the underlying symbolic knowledge, for instance:

“IF patient has a fever AND patient has a cough AND patient has difficulty breathing THEN patient may have pneumonia”.

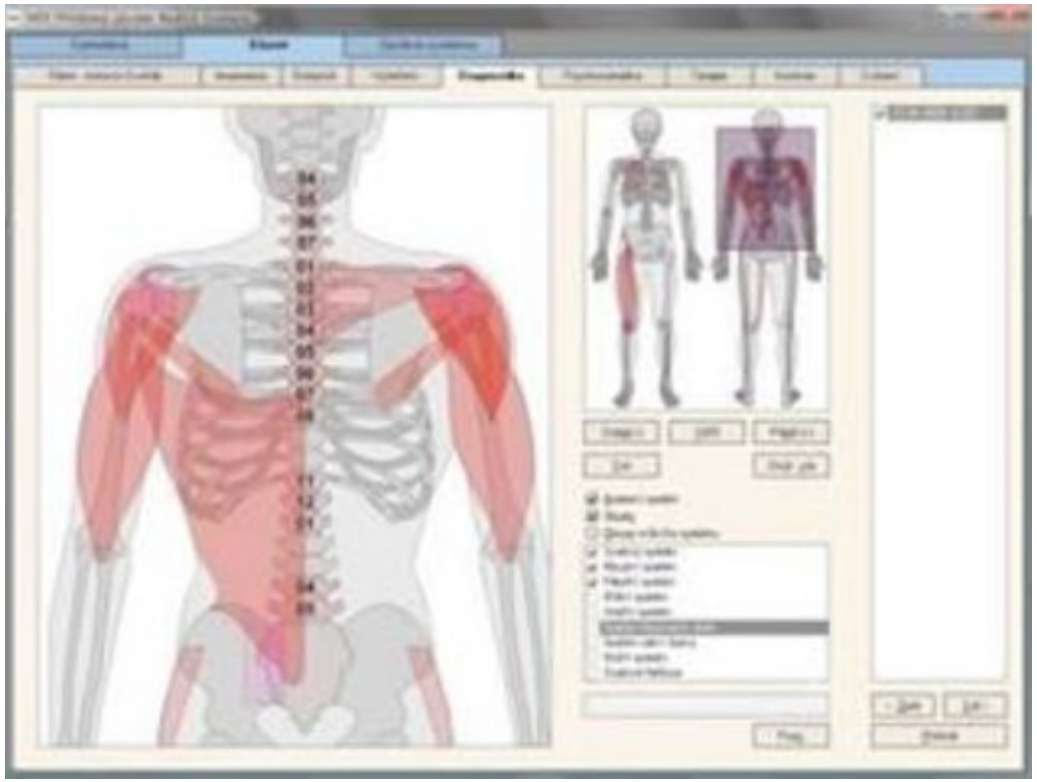


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Symbolic AI: examples
Natural language processing

In natural language processing (NLP) – *the subfield of AI concerned with processing data encoded in natural language* – symbolic AI systems use symbols to represent grammar, syntax, language semantic rules, and so on.

Again, understanding and/or generating natural language is performed by reasoning over this symbolic knowledge.

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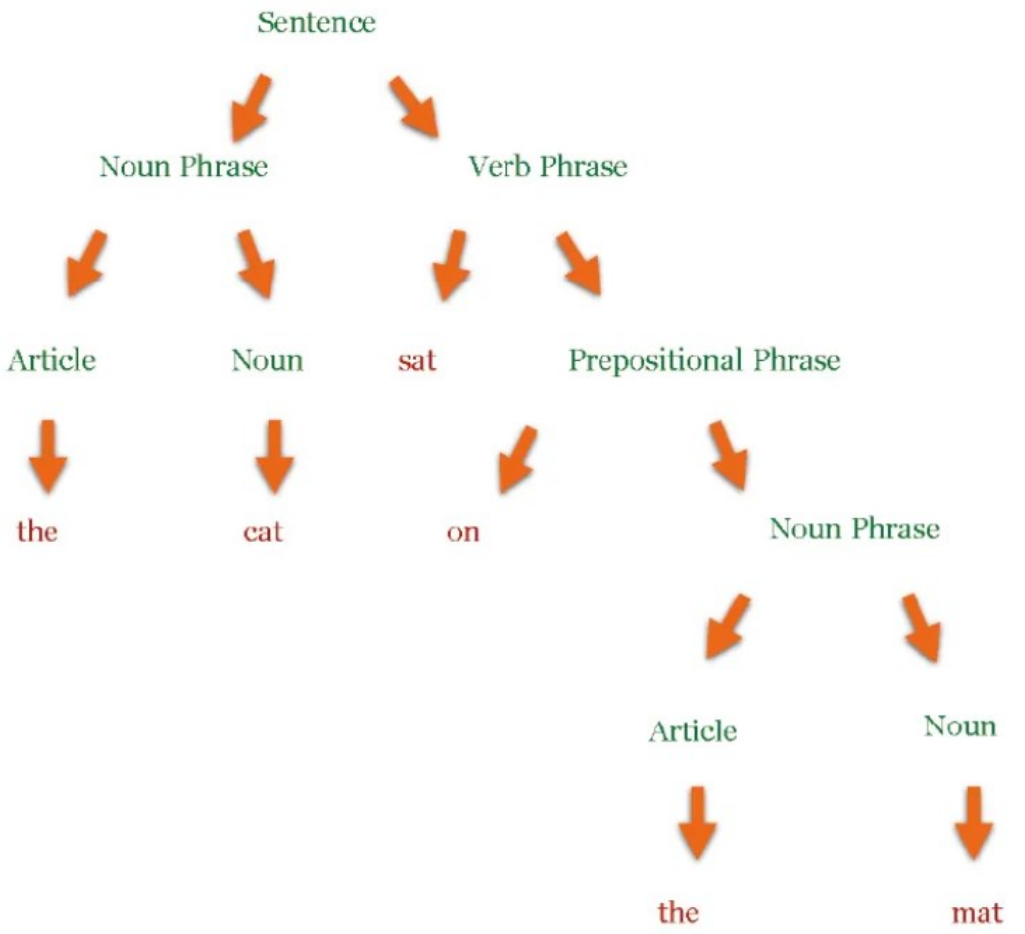


Fig 1.2 An example derivation of the sentence "the cat sat on the mat"

Symbolic AI: techniques, tools, systems

Symbolic AI is mainly based on knowledge representation (through formal symbols) and reasoning over such a knowledge.

Logic programming is a predominant tool used in symbolic AI to perform knowledge representation and reasoning.

Another prominent technique of symbolic AI is problem solving through search:

- Represent the target problem as a tree (or graph) depicting all the possible states of the problem.
- Select the best choice by searching for the "most favorable" state within such a search tree (i.e., a state that makes the player win the game).
- Example: in game playing (e.g., chess), the search tree is given by all the possible configurations of the target game (e.g., all the possible configurations of the chessboard), where the father-child relationships of the tree are given by the dependencies between one configuration and another based on a specific choice performed by the players of the game; the goal of the search is to select at every time the choice that has the highest chance to lead to a victorious state within the search tree.

Systems that typically make use of symbolic AI include expert systems, rule-based systems, inference engines.

Symbolic AI and GOFAI

Symbolic AI is also known as "Good old-fashioned AI" (GOFAI), because the principles and techniques at the basis of symbolic AI are the ones that have been traditionally employed at the dawn of AI.

The "GOFAI" term was coined by philosopher John Haugeland, in 1985.

At that time, the "old-fashioned" term had a negative connotation, as it was used to imply that this type of AI had been superseded by the emerging machine-learning, neural-network-based paradigm.

Nowadays, the "GOFAI" term has no negative connotation at all!

It just denotes a specific form of AI.

Symbolic AI nowadays

Siri and other digital assistants use (among others) symbolic AI to understand natural language and provide responses.

Medical diagnosis systems use symbolic AI to provide recommendations to doctors based on patient symptoms.

Autonomous cars use (among others) symbolic AI to make decisions based on the environment, such as recognizing stop signs and traffic lights.

Computer vision systems use symbolic AI to recognize objects and patterns in images.

Symbolic AI: pros and cons

Main pros:

explicit knowledge representation in a formal and structured way; reasoning based on formal tools and techniques; transparency and interpretability; flexibility in changes of rules and knowledge; no need for vast amount of examples to derive knowledge or rules.

Main cons:

incomplete knowledge; difficulty with handling uncertain or ambiguous information (it relies on precise and unambiguous representations of knowledge); limited scalability; limited ability to learn and adapt.

Forms of AI:
Machine learning and Connectionist AI

Machine learning

Machine learning is the field of study (subfield of AI) which focuses on computing systems capable of learning from examples, building hypotheses about the world based on those examples, and use such hypotheses as mathematical models to make predictions on or infer properties of unseen data.

Example:

- Collect examples of emails, along with the information whether they are spam or not.
- [Training] Use a machine learning method to build a mathematical model (basically, a function) from those example emails which derives the characteristics of an email for being spam or not.
- [Inference] Use such a mathematical model to state whether any future, previously unseen email is spam or not.

Machine learning and AI

In the public eye, there is sometimes confusion between the terms “artificial intelligence” and “machine learning”.

Machine learning is a subfield of AI that studies the ability to improve performance based on experience. It is a specific form of AI.

Some AI systems use machine learning methods to achieve competence, but some do not.

Machine learning vs. symbolic AI

Machine learning can be viewed as a "bottom-up" approach to AI, where intelligence is built going from particular (examples) to general (model used to future inference/prediction).

It is the dual of symbolic AI, which is instead an inherently "top-down" approach to AI, where intelligence is built going from general (initial hypotheses, e.g., statements in the knowledge base), to particular (further conclusions, derived via reasoning on the initial hypotheses).

Machine learning and artificial neural networks

There exist a plethora of different techniques to build a machine learning model.

Perhaps today's most popular machine learning technique is artificial neural network (or, simply neural network).

- This has been determined by the resurge of interest in neural networks since the early 2010s, due to important progresses, both technological (e.g., the advent of more powerful or dedicated hardware such as GPUs, TPUs) and conceptual (e.g., the progresses in the big data field or the definition of more powerful neural-network architectures such as Transformer).

Neural networks are so popular in the machine learning and AI landscape that, in today's public eye, typically the term "neural network" is viewed as a synonym of "machine learning" (or even as a synonym of "AI" in general).

- This is of course simplistic and incorrect, as neural networks are just one particular technique to achieve machine learning and AI, but absolutely not the only one!

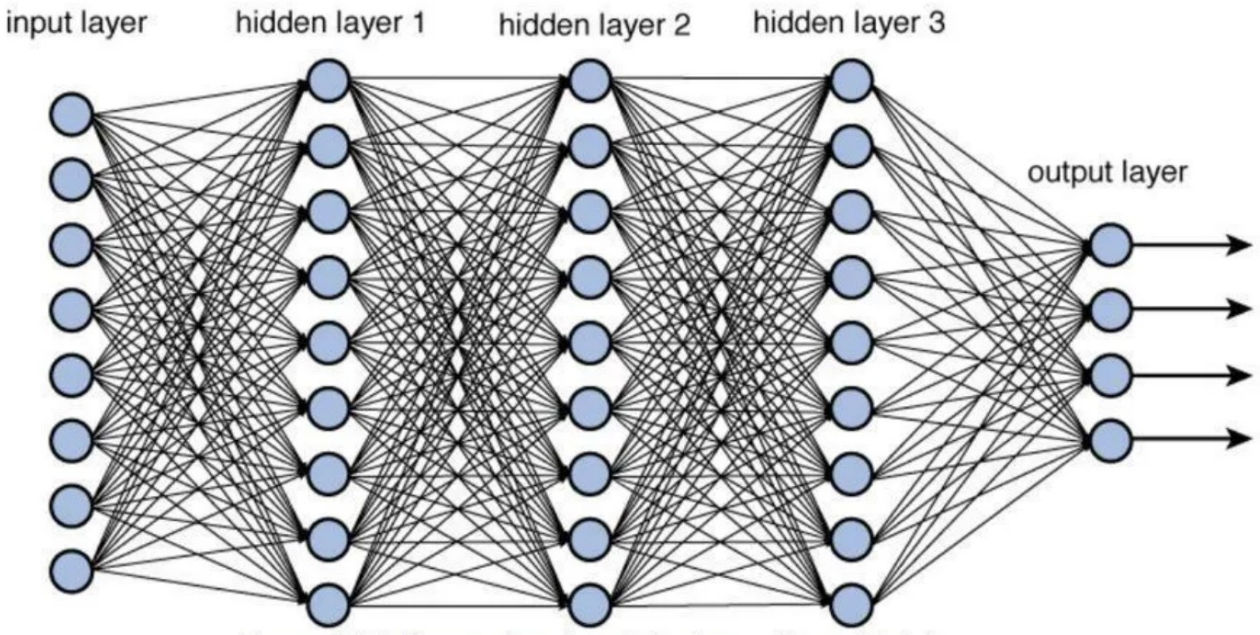
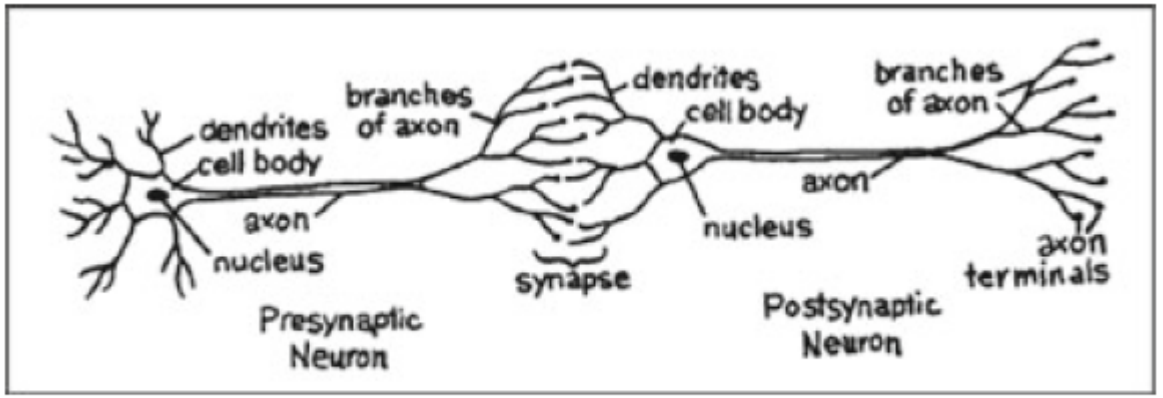


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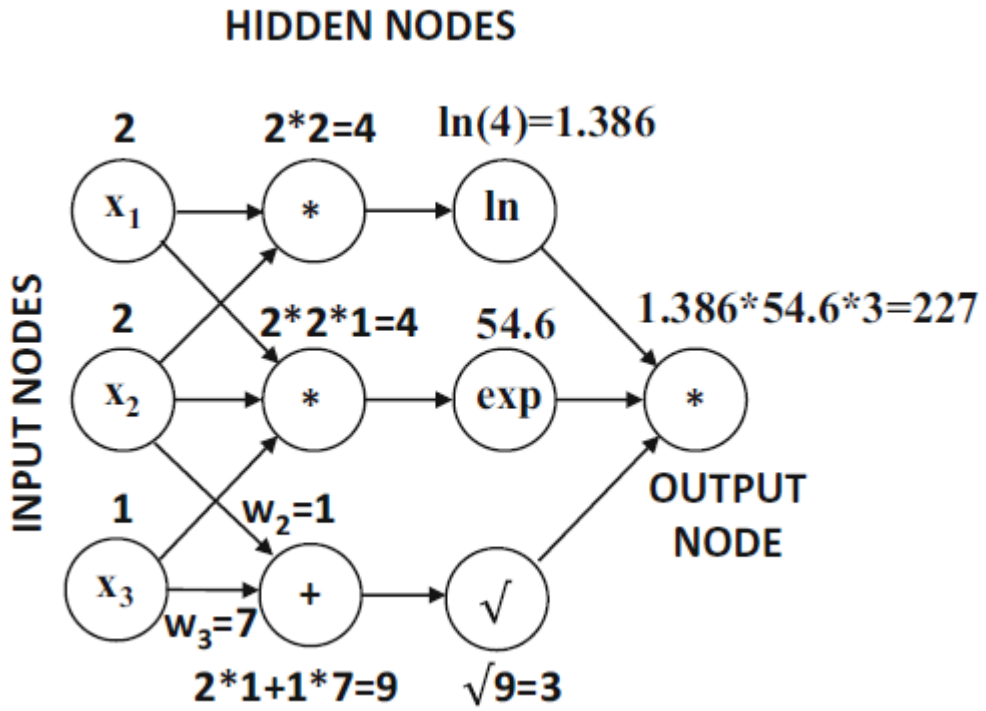
Neural networks

image source



(a) Biological neural network

image source



(b) Artificial neural network

Neural networks

Neural networks are a machine learning technique that is inspired by the structure and function of biological neural networks in animal brains.

A neural network consists of connected units or nodes called artificial neurons, which loosely model the neurons in the brain. These are connected by edges, which model the synapses in the brain.

Each artificial neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons.

The "signal" is simply a real number, and the output of each neuron is computed by some non-linear function of the sum of its inputs, called the activation function.

The strength of the signal at each connection is determined by a weight, which adjusts during the learning process from the available examples.

- Therefore, the learning process (training) within a neural network consists of determining the right weigths to be assigned to every connection.

Neural networks

How neural networks might recognize an image depicting an apple:

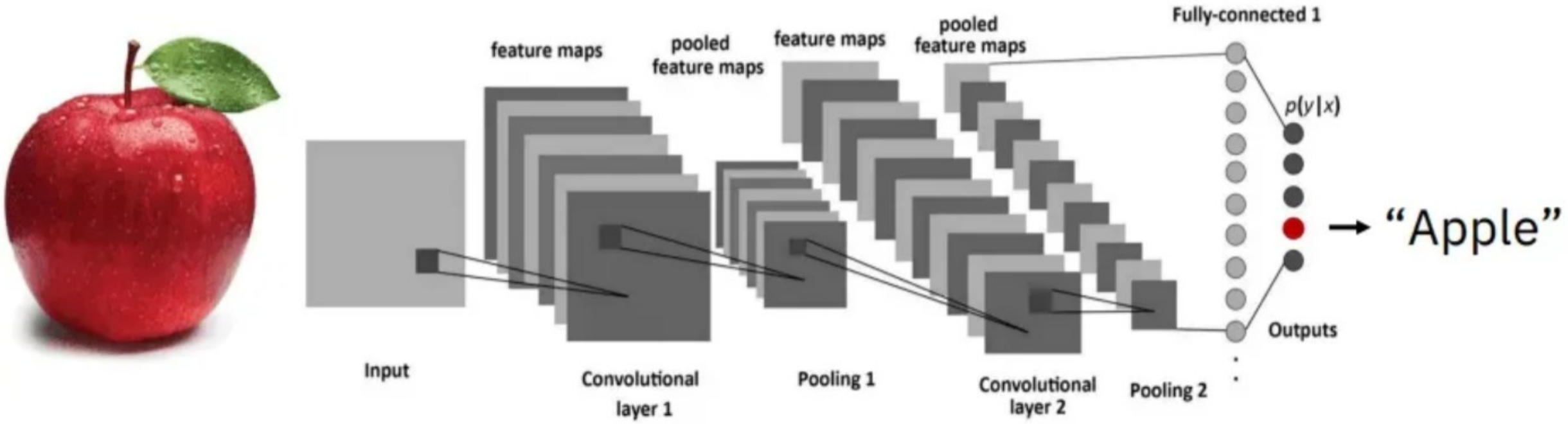


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Deep learning

The term deep learning refers to machine learning using many layers of simple, adjustable computing elements.

In the context of neural networks, this corresponds to having many layers of connected artificial neurons.

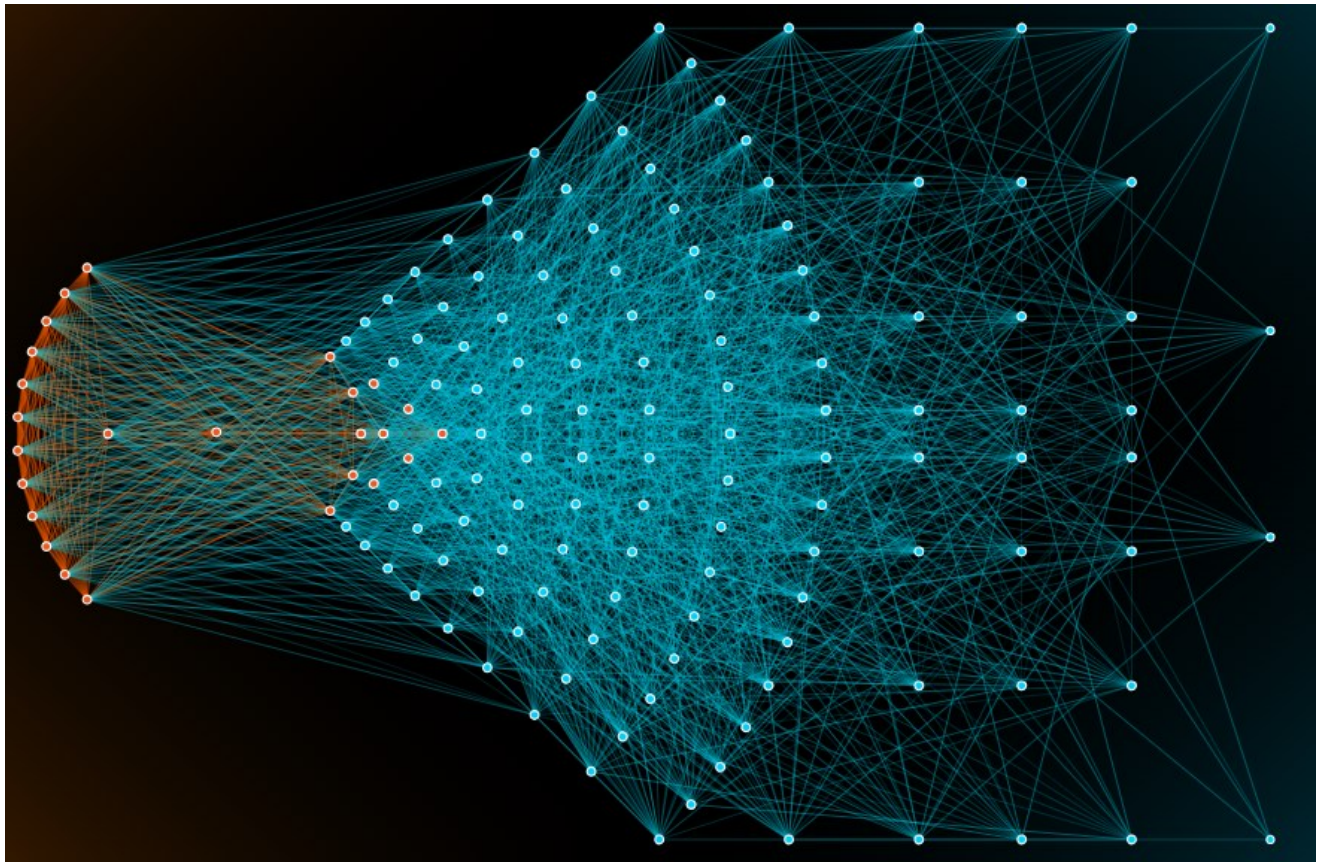


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Connectionist AI

Neural networks are models that basically consist of interconnections among simple processing units (i.e., artificial neurons).

This has led to coining the term "connectionist AI" and to let it correspond to a form of AI based on machine learning and, specifically, on neural network.

Symbolic AI and connectionist AI are perhaps today's main schools of thought in AI, and the dualism between these two schools is perhaps the most important and critical one in the whole AI landscape.

Connectionist AI: pros and cons

Main pros:

ability to learn and adapt; ability to handling uncertain or ambiguous information; capability of handling large amount of data.

Main cons:

knowledge is not symbolic (it is “encoded” into connections between neurons), thus non-human-interpretable; the output outcomes are non-human-interpretable either.

Forms of AI: Neuro-symbolic AI

Symbolic AI and connectionist AI

As said, AI has long had a split between symbolic systems (based on logical and probabilistic inference) and connectionist systems (based interconnections among simple processing units).

Symbolic AI and Connectionist AI represent two distinct yet complementary approaches to AI, with their own pros and cons.

- Symbolic AI systems allow us to string together long chains of reasoning and to take advantage of the expressive power of structured representations.
- Connectionist AI systems can adapt to unseen situations and recognize patterns even in the face of noise and uncertainty.

Symbolic AI excels in tasks that require explicit knowledge, logical reasoning , and interpretability.

Connectionist AI shines in pattern recognition, adaptive learning, and handling large amounts of data.

Neuro-symbolic AI

A continuing challenge for AI is to bring the symbolic AI and connectionist AI paradigms together.

- This is the main goal of the so-called “neuro-symbolic AI”.

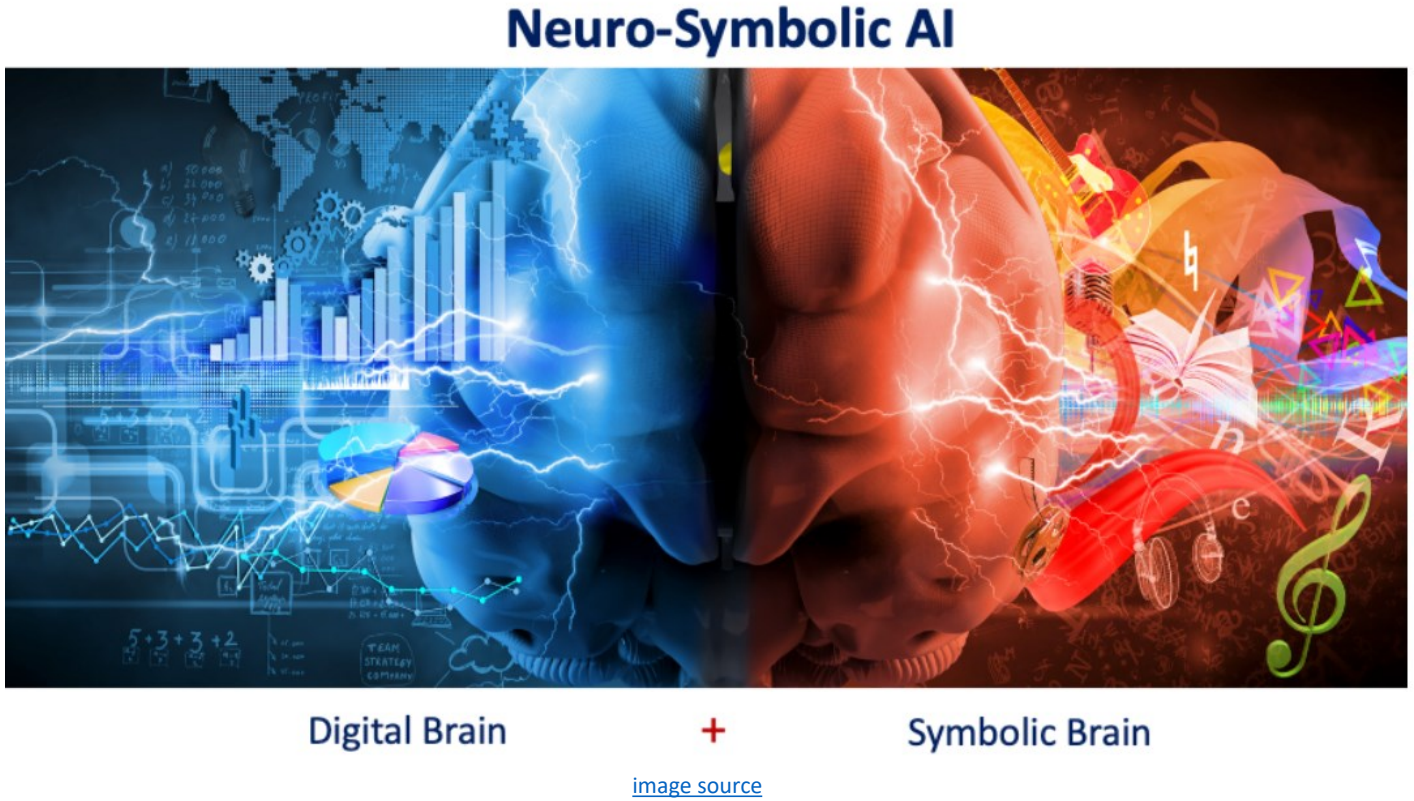


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Neuro-symbolic AI: benefits

The potential of neuro-Symbolic AI is vast.

By combining learning and reasoning, these systems could potentially understand and interact with the world in a way that is much closer to how humans do.

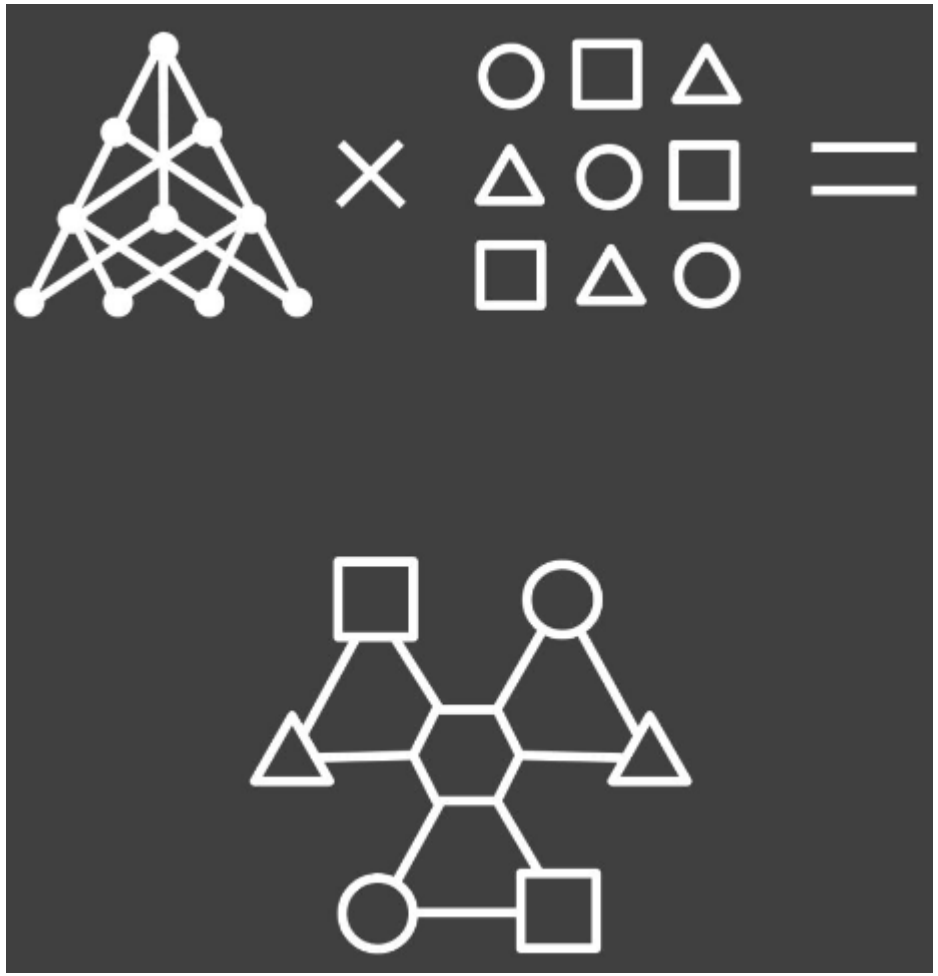
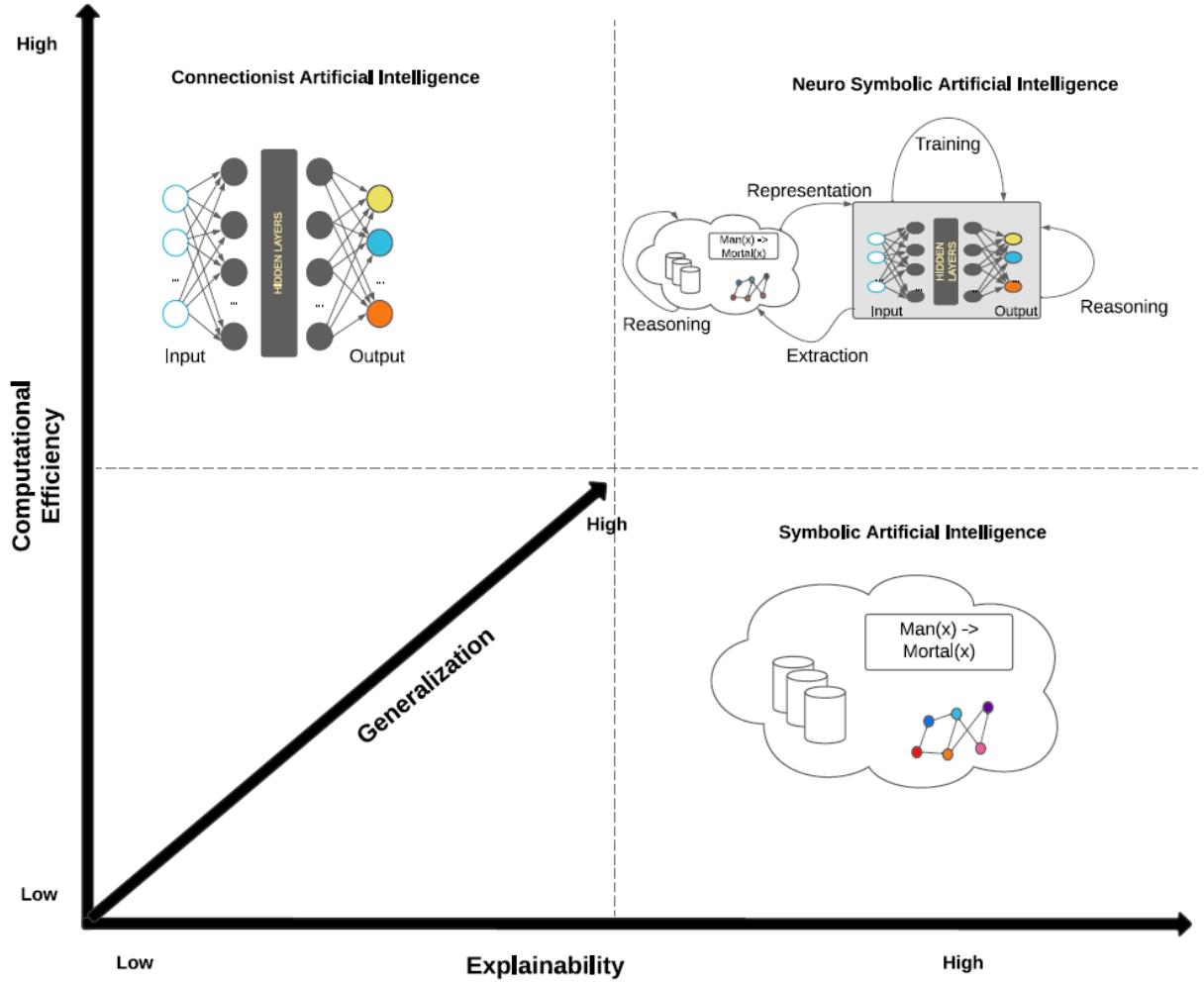


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Neuro-symbolic AI: benefits

Fig. 1 The drawbacks of both the fields individually in terms of 'Explainability', 'Efficiency', and 'Generalization', when the fields merge together to form neuro-symbolic artificial intelligence, all three characteristics are high.

Source: Bhuyan et al., https://doi.org/10.1007/s00521-024-09960-z



Neuro-symbolic AI applications: self-driving cars

Self-driving cars need to:

- recognize objects (cars, pedestrians, traffic signs, etc.) – a task well-suited for neural networks;
- make decisions based on these recognitions and according to traffic rules – a task well-suited for symbolic AI.

A neuro-Symbolic AI system in this context would use:

- a neural network to learn to recognize objects from data (images from the car's cameras), and
- a symbolic AI system to reason about these objects and make decisions according to traffic rules.

This combination allows self-driving cars to interact with the world in a more human-like way, understanding the context and making reasoned decisions.

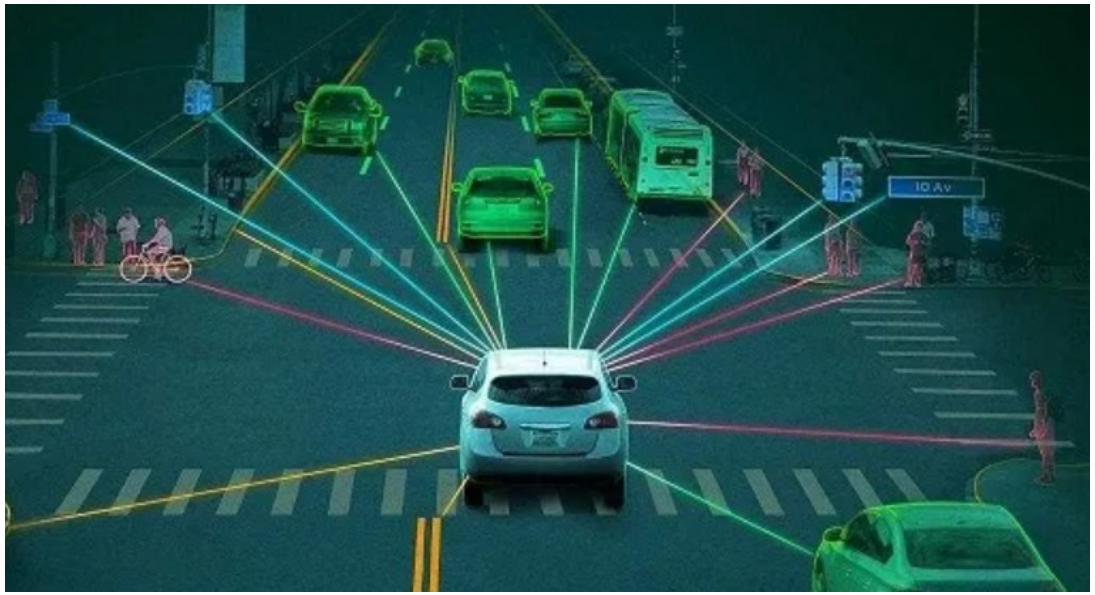


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Neuro-symbolic AI applications: self-driving cars

Neuro-symbolic AI systems for self-driving cars are already available nowadays.

For instance, Sharifi et al.'s *"general neuro-symbolic method for online visual sensemaking using answer set programming (ASP)"*, which is *"systematically formalised and fully implemented"*.

- https://arxiv.org/abs/2307.01316v2

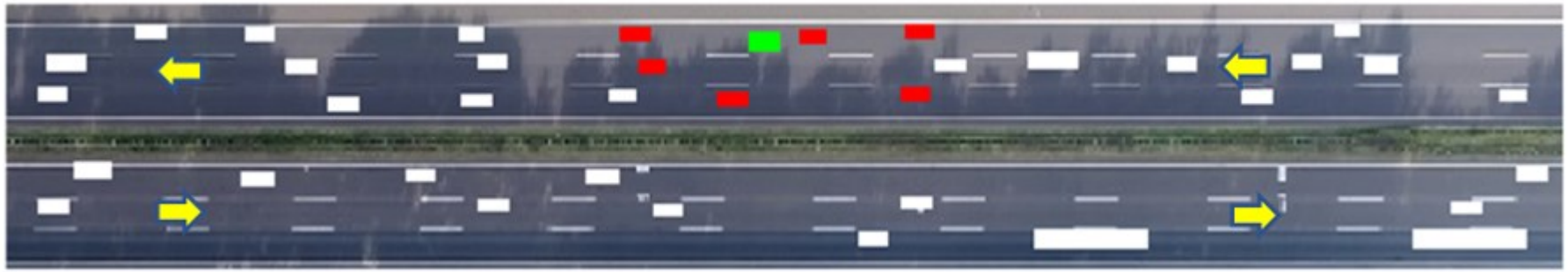


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Neuro-symbolic AI applications: solving math problems

Google's DeepMind has developed a system called AlphaGeometry, which is capable of solving geometry problems at the level of the brightest high-school students. This system pairs a language model with a symbolic deduction engine, forming a neuro-symbolic system that mimics how humans solve mathematical problems.

DeepMind researchers explained that they tested AlphaGeometry on 30 geometry problems at the difficulty level of the International Mathematical Olympiad, in which the brightest high school students compete. It completed 25 of the problems within the time limit, beating the previous record of 10. This performance is comparable with that of the average International Mathematical Olympiad gold medallist from 2000 to 2022 (25.9).

- AlphaGeometry is described in a paper published in Nature in Jan 2024 (https://www.nature.com/articles/s41586-023-06747-5)

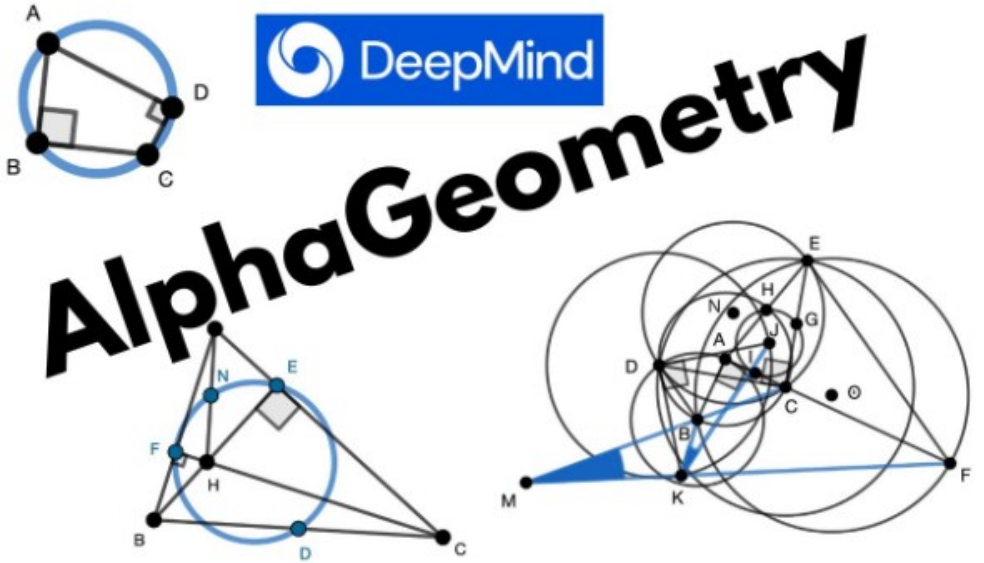


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Neuro-symbolic AI: approaches

Neuro-symbolic AI methods can be classified in two main categories:

1. Methods that "inject" symbolic knowledge into neural networks.

Methods of this kind compress structured symbolic knowledge and produce representations of such a knowledge that are suitable to be processed by neural networks. The compressed, neural-network-processable knowledge can be integrate with neural patterns, in order to have neural networks operate by exploiting the resulting integrated neural patterns

2. Methods that "incorporate" neural patterns into symbolic knowledge.

Method of this kind extract information from neural patterns to allow for mapping to structured symbolic knowledge (lifting) and perform symbolic reasoning over a knowledge that is augmented by neural patterns.

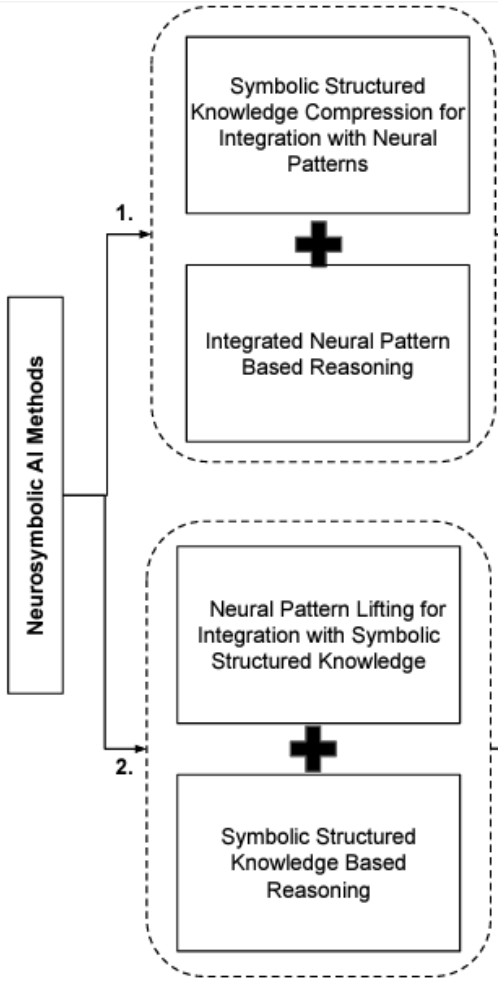


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Forms of AI: Agent-centric AI

Intelligent agents

Earlier AI approaches to action tended to focus on single, isolated software systems that acted in a relatively inflexible way, automatically following pre-set rules.

Modern technologies and software applications have created a need for artificial entities that are more autonomous, flexible, and adaptive, and that operate as social entities by interacting with each other and with the humans or the environment.

- Such artificial entities are termed “intelligent agents” or simply “agents”

Intelligent agents

The concept of agent is used often in AI, when a sequence of decisions ("actions") need to be made in order to achieve a particular goal.

An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators:

- A human agent has eyes, ears, and other organs for sensors and hands, legs, vocal tract, and so on for actuators.
- A robotic agent might have cameras and infrared range finders for sensors and various motors for actuators.
- A software agent receives file contents, network packets, and human input (keyboard/mouse/touchscreen/voice) as sensory inputs and acts on the environment by writing files, sending network packets, and displaying information or generating sounds.

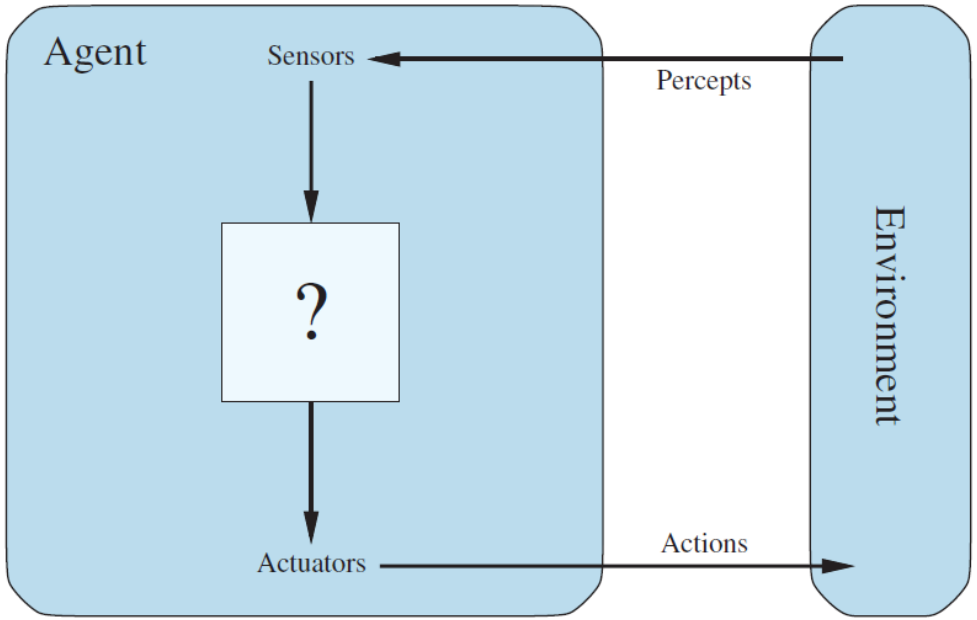


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Intelligent agents: environment

The environment which intelligent agents interact with could be everything—even the entire universe!

In practice it is just that part of the universe whose state we care about when designing this agent: the part that affects what the agent perceives and that is affected by the agent's actions.

The term "percept" is used to refer to the content an agent's sensors are perceiving.

An agent's percept sequence is the complete history of everything the agent has ever perceived.

In general, an agent's choice of action at any given instant can depend on its built-in knowledge and on the entire percept sequence observed to date, but not on anything it hasn't perceived.

Intelligent agents: examples

Tasks such as playing a game of chess, performing a robotic task, finding a path in a maze, replying to natural questions posed by a human, all require a sequence of decisions from an intelligent agent.

An example of an agent might be a robot, a chatbot, a chessplaying entity, or a route planner.

Intelligent agents: examples

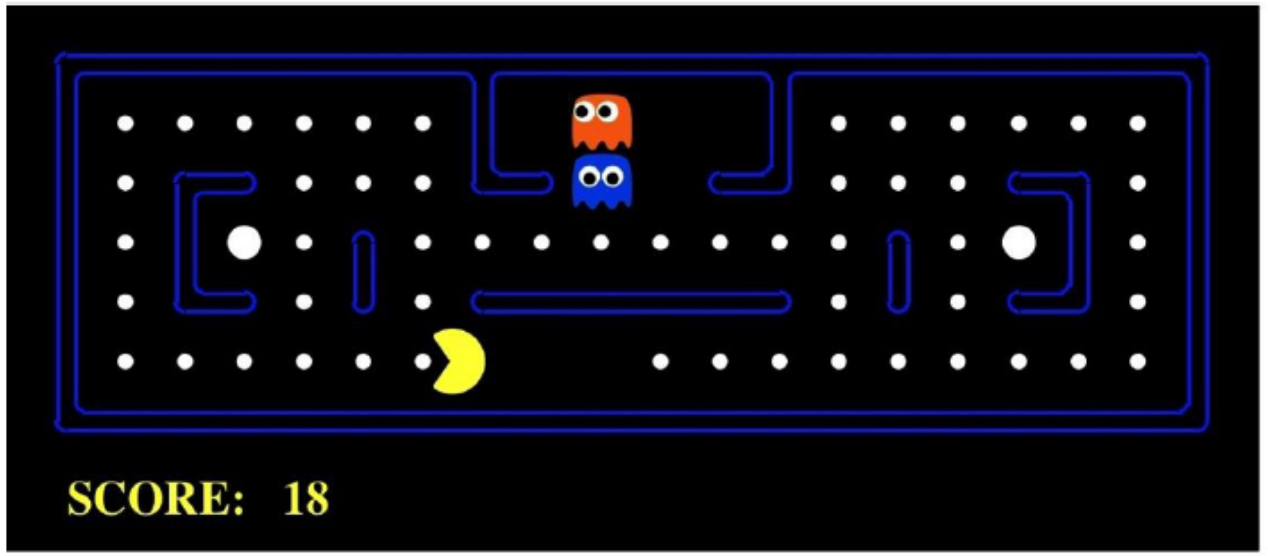
Table 1.2: Agent examples

Agent	Sensor	Actuator	Objective	State	Environment
Cleaning Robot	Camera Joint sensor	Limbs Joints	Cleanliness evaluation	Object/joint positions	Cleaned space
Chess agent	Board input interface	Move output interface	Position evaluation	Chess position	Chess board
Self-driving car	Camera Sound sensor	Car control interface	Driving safety and goals	Speed/Position in traffic	Traffic conditions
Chatbot	Keyboard	Screen	Chat evaluation	Dialog history	Chat participants

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Intelligent agents: examples

Pac-Man is a registered trademark of Namco-Bandai Games, used here for educational purposes



Pac-Man is an agent!

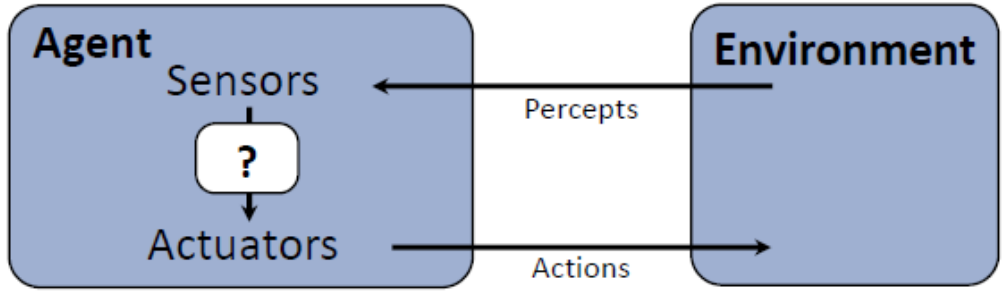


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Agent-based AI tasks

It is only in the case of problems associated with a sequence of decisions that it becomes important to use the concept of agent and a state of the system.

Such settings are referred to as sequential.

An intelligent agent plays the same role that a human plays in a biological system, based on a sequence of choices that achieve a desired goal.

Non-agent-based AI tasks

One does not always encounter the notion of an agent in all forms of AI.

For example, the concept of agent rarely arises for single-decision problems in machine learning; an example is that of classification, where one is trying to categorize an instance into one of many classes (e.g., categorize whether an image contains animals or not).

Such environmental settings are referred to as episodic, and even though the notion of agent is implicit, it is rarely used in practice.

In these cases, the solution to this problem is a single-step process, and therefore the question of system state or environment is not as important as it would be in a sequential process where an action affects subsequent states.

Therefore, in such cases, the implicit agent receives a single percept and performs a single action. The next percept-action pair is independent of the current one. In other words, the individual episodes are independent of one another.

AI goals and subfields

Subfields of AI

The general problem of simulating (or creating) intelligence has traditionally been broken into subproblems.

These consist of particular traits or capabilities that researchers expect an intelligent system to display.

Any subproblem typically lead to a specific subfield of AI.

The most prominent AI subfields include (but they are not limited to):

- problem solving, knowledge representation, reasoning, planning, machine learning, machine perception, intelligent agents, robotics, natural language processing, computer vision, social intelligence (e.g., affective computing), cognitive intelligence, generative AI

Subfields of AI

Problem solving:

Enabling machines solving complex problems (e.g., games, puzzles).

Knowledge representation:

Representing knowledge in ad-hoc formats that can allow machines exploit it for intelligent tasks.

(Automated) Reasoning:

Generating intelligent outcomes and making inference from available knowledge.

(Automated) Planning and scheduling:

Generating sequences of actions or strategies to achieve specific goals.

Subfields of AI

Machine learning:

Learning from examples and generalizing to unseen data and situations.

Machine perception:

Ability to use input from sensors (e.g., cameras, microphones, wireless signals, active lidar, sonar, radar, tactile sensors, and so on) to deduce aspects of the world.

Intelligent agents:

Dealing with entities (“agents”) that perceive and takes actions in the world.

Robotics:

Designing and building robots.

Subfields of AI

Natural language processing:

"Processing" (i.e., understanding, analyzing, generating, and so on) data encoded in natural (human) language.

Computer vision:

Acquiring, processing, analyzing and understanding digital images.

Social intelligence (e.g., affective computing):

Recognizing, interpreting, processing, simulating human feeling, emotion, and mood.

Cognitive intelligence/cognitive computing:

Simulating human thought processes by mimicking the way how human brains work. It involves machines that can learn, reason, and understand language in a way that is similar to how humans do.

Subfields of AI

Generative AI:

Generating digital content of various types: text, images, videos, music, and so on.

The recent progresses made in generative AI (adopted in commercial tools such as, e.g., the popular ChatGPT) has been one of the key factors for the recent hype in AI.

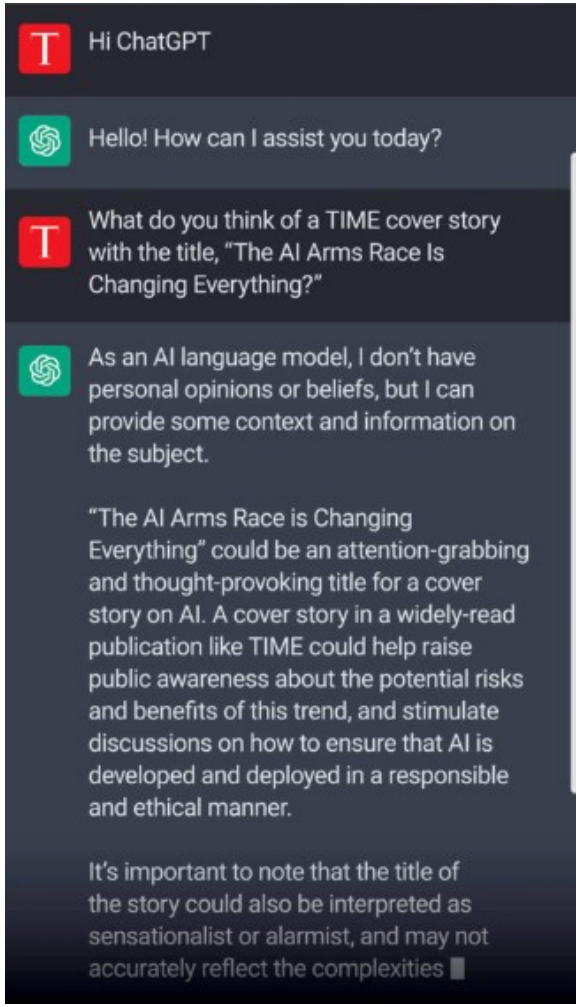


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Ethics of AI

Benefits of AI

Given that AI is a powerful technology, we have a moral obligation to use it well, to promote the positive aspects and avoid or mitigate the negative ones. The positive aspects are many:

AI can save lives through:

- improved medical diagnosis,
- new medical discoveries,
- better prediction of extreme weather events,
- safer driving with driver assistance.

There are also many opportunities for AI to improve lives:

- Microsoft's AI for Humanitarian Action program applies AI to recovering from natural disasters, addressing the needs of children, protecting refugees, and promoting human rights.
- Google's AI for Social Good program supports work on rainforest protection, human rights jurisprudence, pollution monitoring, measurement of fossil fuel emissions, crisis counseling, news fact checking, suicide prevention, recycling, and other issues.
- The University of Chicago's Center for Data Science for Social Good applies AI to problems in criminal justice, economic development, education, public health, energy, and environment.

Benefits of AI

Further key benefic AI applications include:

- AI applications in crop management and food production help feed the world.
- Optimization of business processes using AI will make businesses more productive, increasing wealth and providing more employment.
- Automation can replace the tedious and dangerous tasks that many workers face, and free them to concentrate on more interesting aspects.
- People with disabilities will benefit from AI-based assistance in seeing, hearing, and mobility.
- Machine translation already allows people from different cultures to communicate.
- Software-based AI solutions have near zero marginal cost of production, and so have the potential to democratize access to advanced technology.

Responsible AI

Despite the many positive aspects of AI, we shouldn't ignore the negative ones, even if they are unintended.

The set of principles, guidelines, and mitigations strategies for the negative impacts of AI is known under the umbrella terms

"Responsible AI", "AI ethics", "Trustworthy AI".

The gist is that every organization that creates AI technology, and everyone in the organization, has a responsibility to make sure the technology contributes to good, not harm.

Responsible AI

The most commonly-required principles for a responsible use of AI are:

- ensure safety;
- establish accountability;
- ensure fairness;
- uphold human rights and values;
- respect privacy;
- reflect diversity/inclusion;
- promote collaboration;
- avoid concentration of power;
- provide transparency;
- acknowledge legal/policy implications;
- limit harmful uses of AI;
- contemplate implications for employment.



image source

AI Act

The European Commission is particularly sensitive to the topic of responsible use of AI, so much that it has provided several initiatives in this regard. Perhaps the most important one of these initiatives is the AI Act.

The AI Act is a European Union regulation concerning AI. It establishes a common regulatory and legal framework for AI within the European Union. It came into force on 1 August 2024, with provisions coming into operation gradually over the following 6 to 36 months.

The AI Act is based on the principle that AI must be developed and used in a safe, ethical and respectful way of fundamental rights and European values.

The AI Act aims to guarantee the safety and ethics of AI, protecting the rights and interests of the people and organizations that are affected by it.

The AI Act provides for a system of sanctions in case of violation of the provisions established. The sanctions may vary depending on the severity and nature of the infringement, and are proportionate to the size and turnover of the responsible economic operator.



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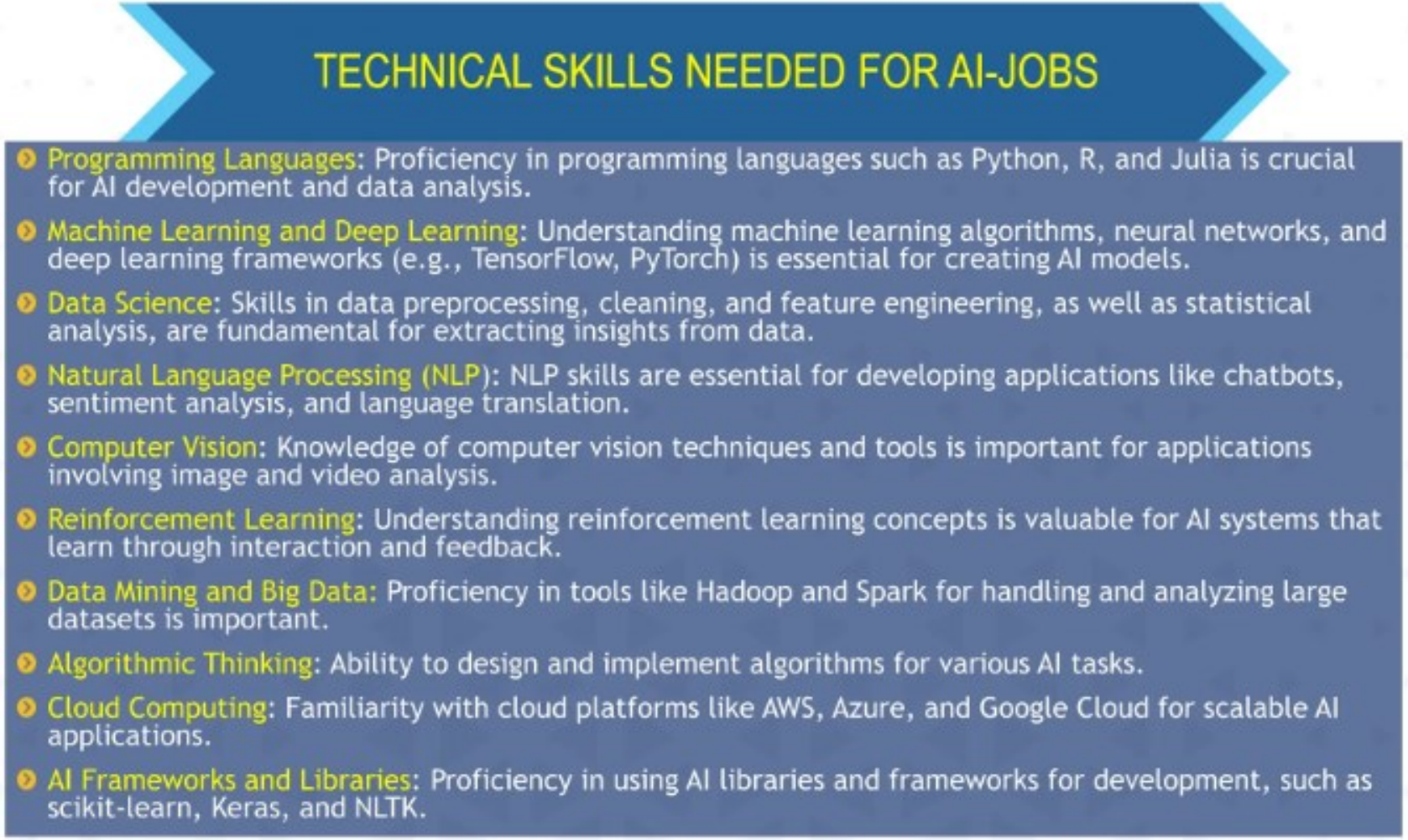
Questions?

AI-driven job market



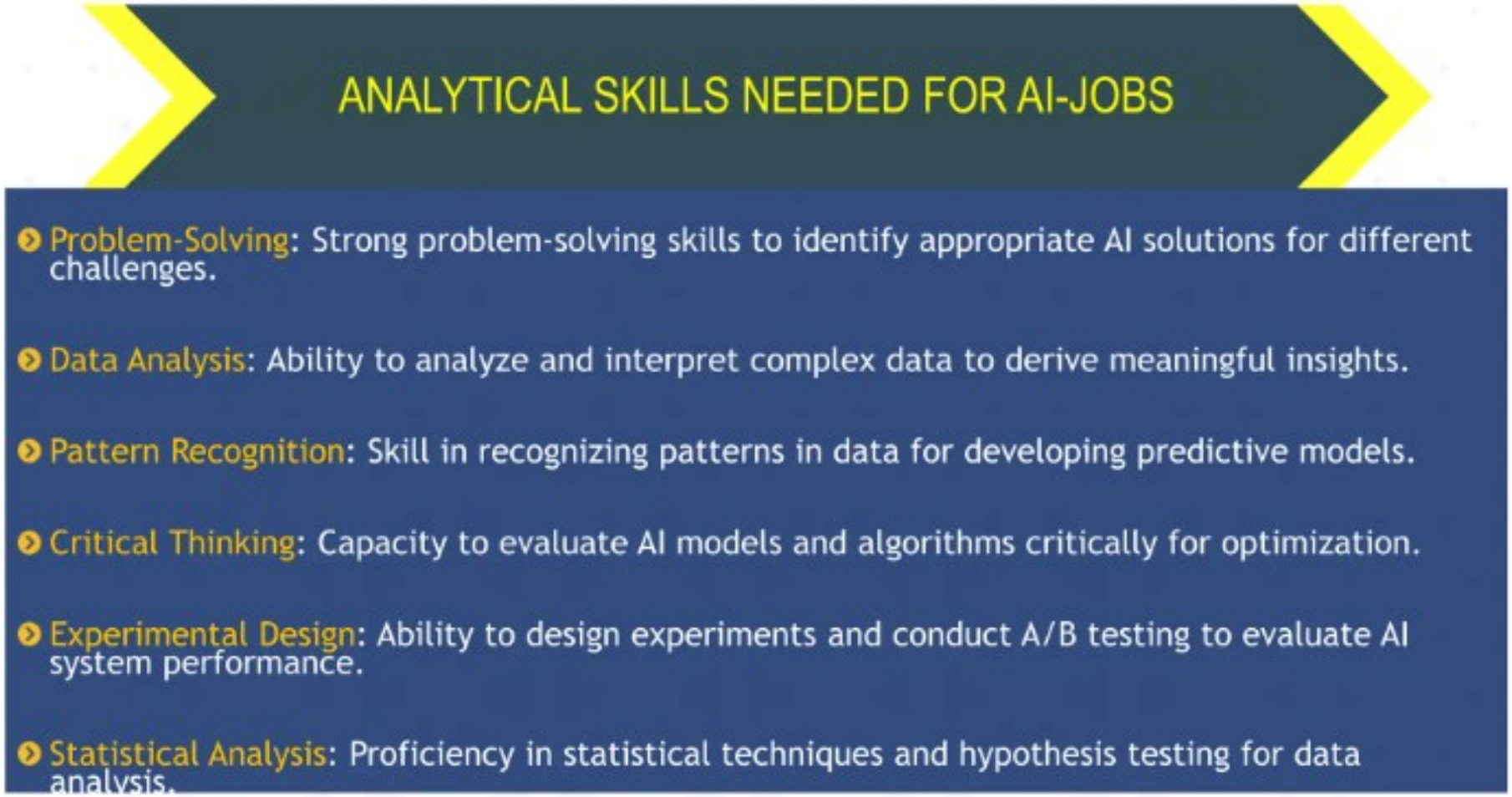
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AI-driven job market



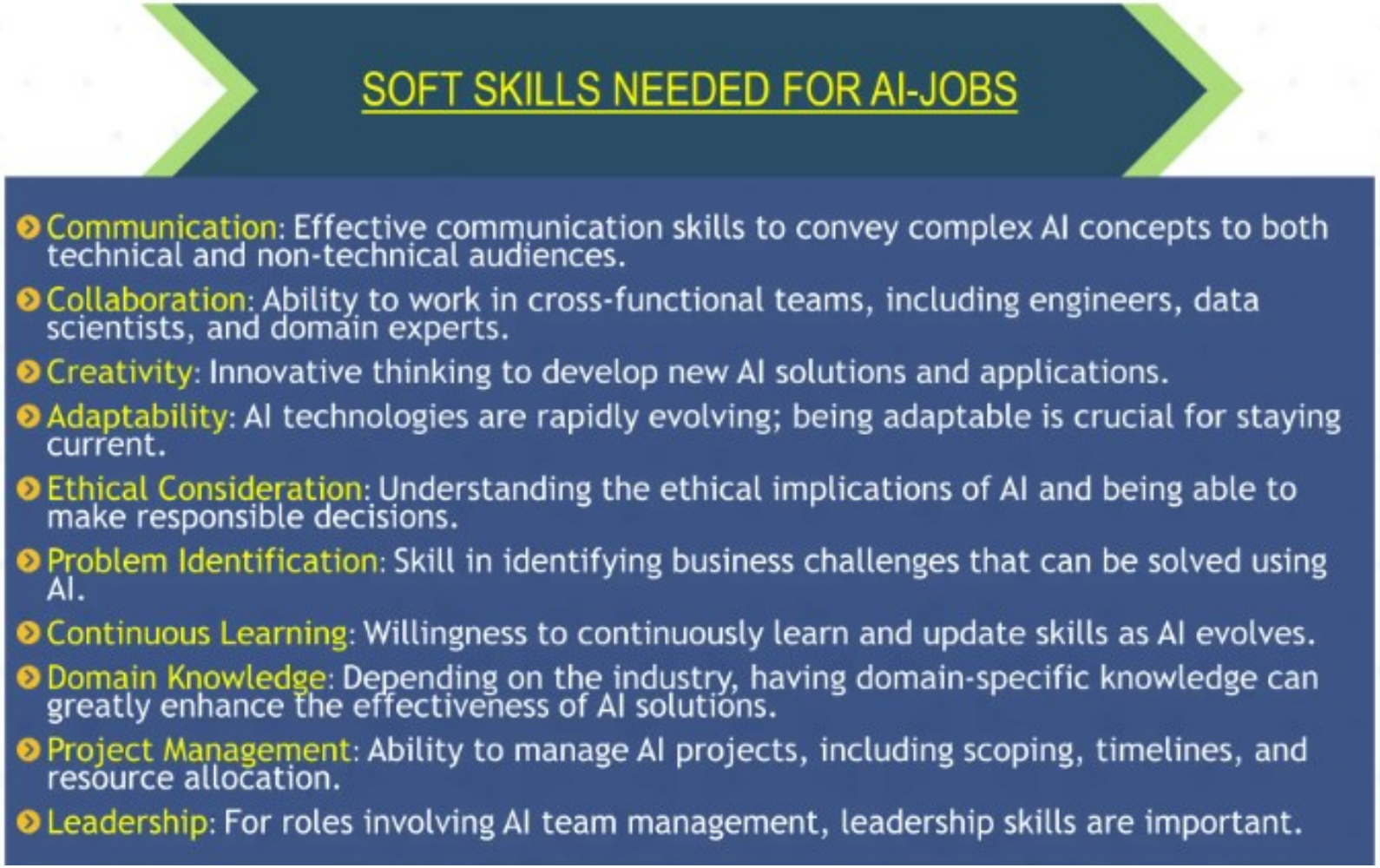
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AI-driven job market



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